

# Cornell Bowers C-IS

College of Computing and Information Science

## Deep Learning

Week [03]: [LLMs]

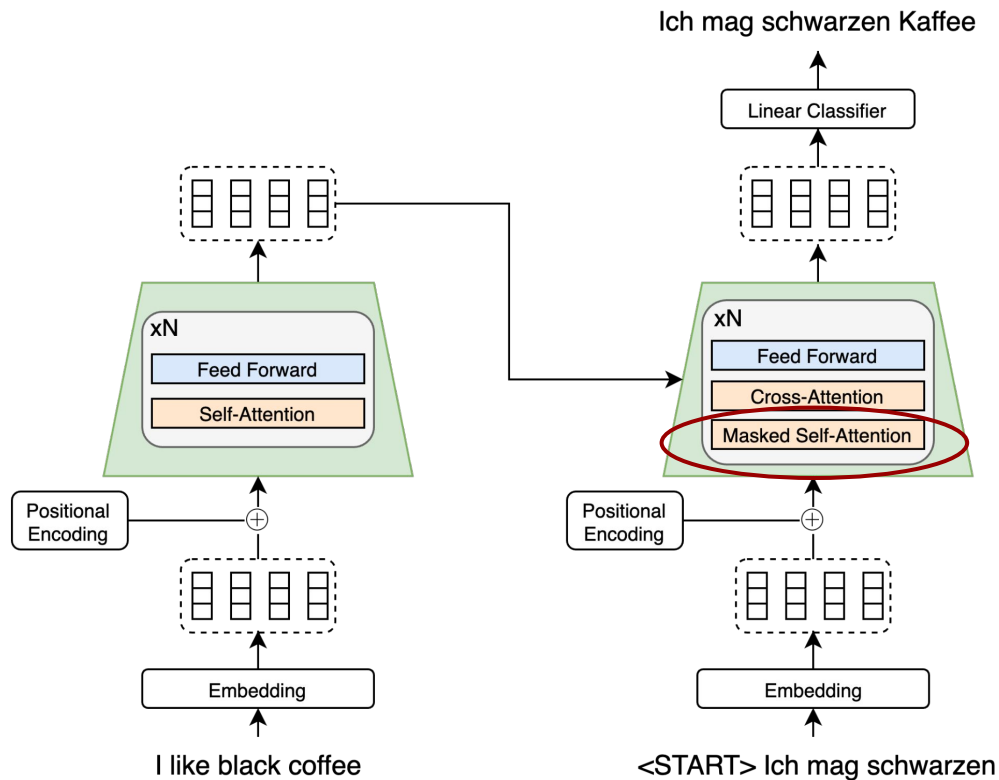
# Thanks to

Varsha Kishore  
Justin Lovelace  
Adam Alnasser  
Elizabeth Kelmenson

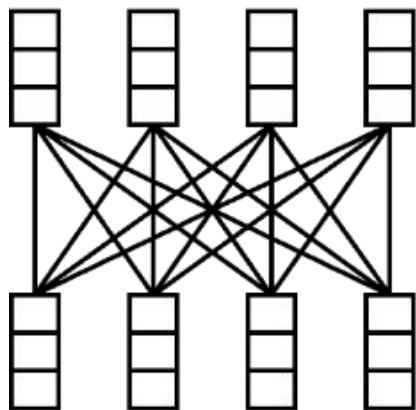
# Logistics

- HW2 due 03/03
- A2 due 03/03
- Project Proposal due 03/07
- HW3 is out 03/03, due 03/12
- A3 is out 03/03, (due TBD but likely 03/22)

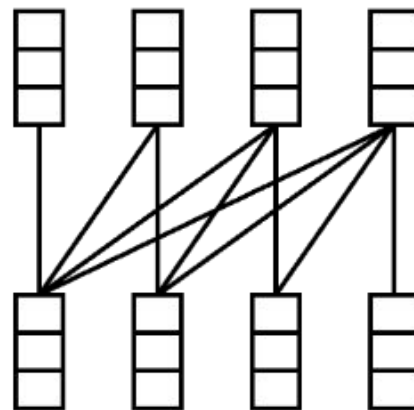
# Transformer Architecture



# Self-Attention vs. Masked Self-Attention



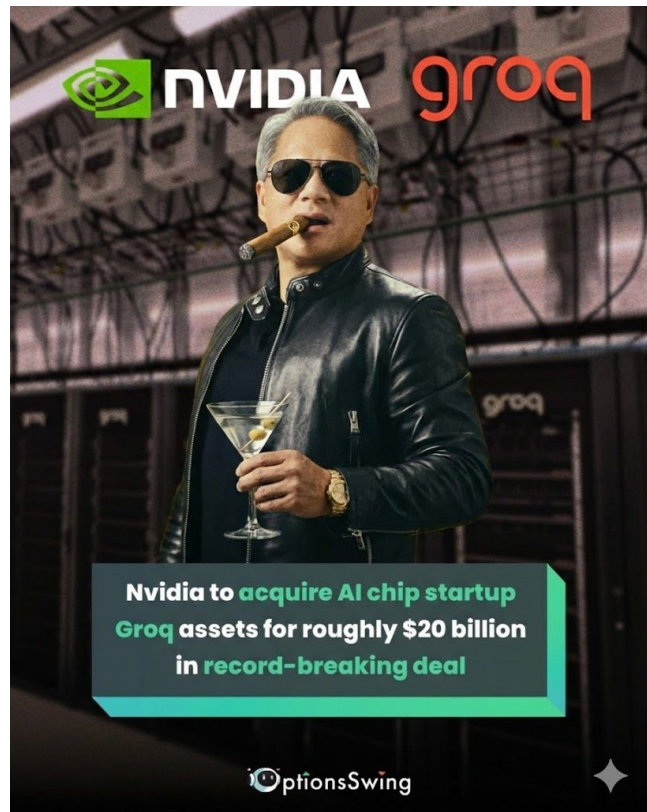
Self-Attention



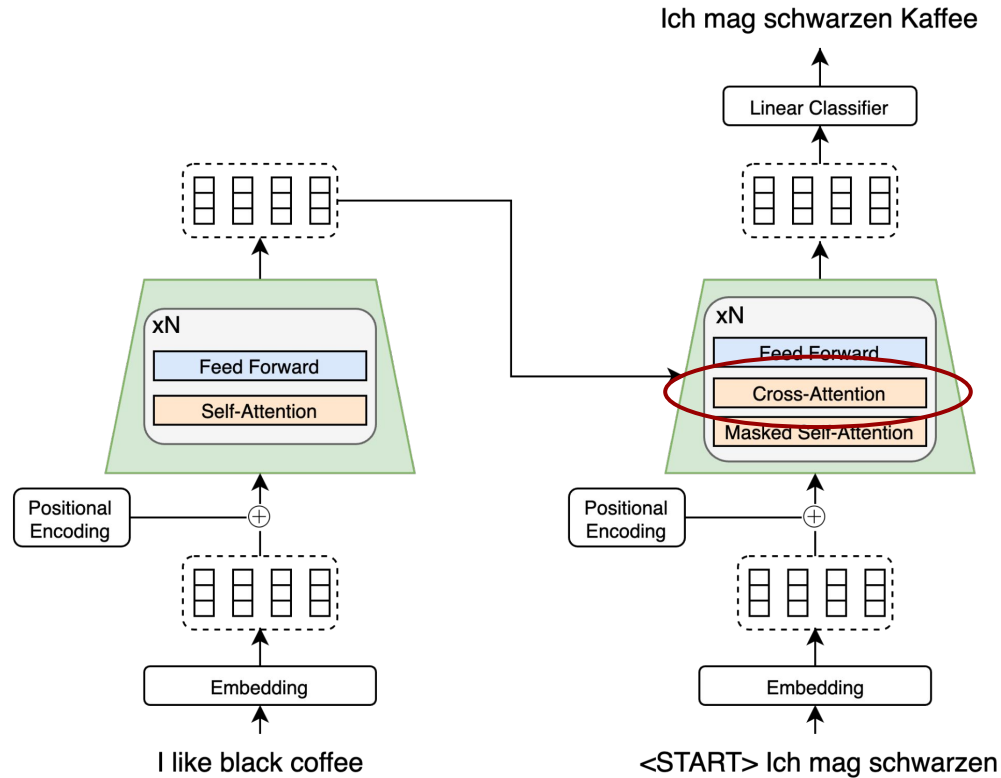
Masked Self-Attention

# What's the issues with Autoregressive Generation?

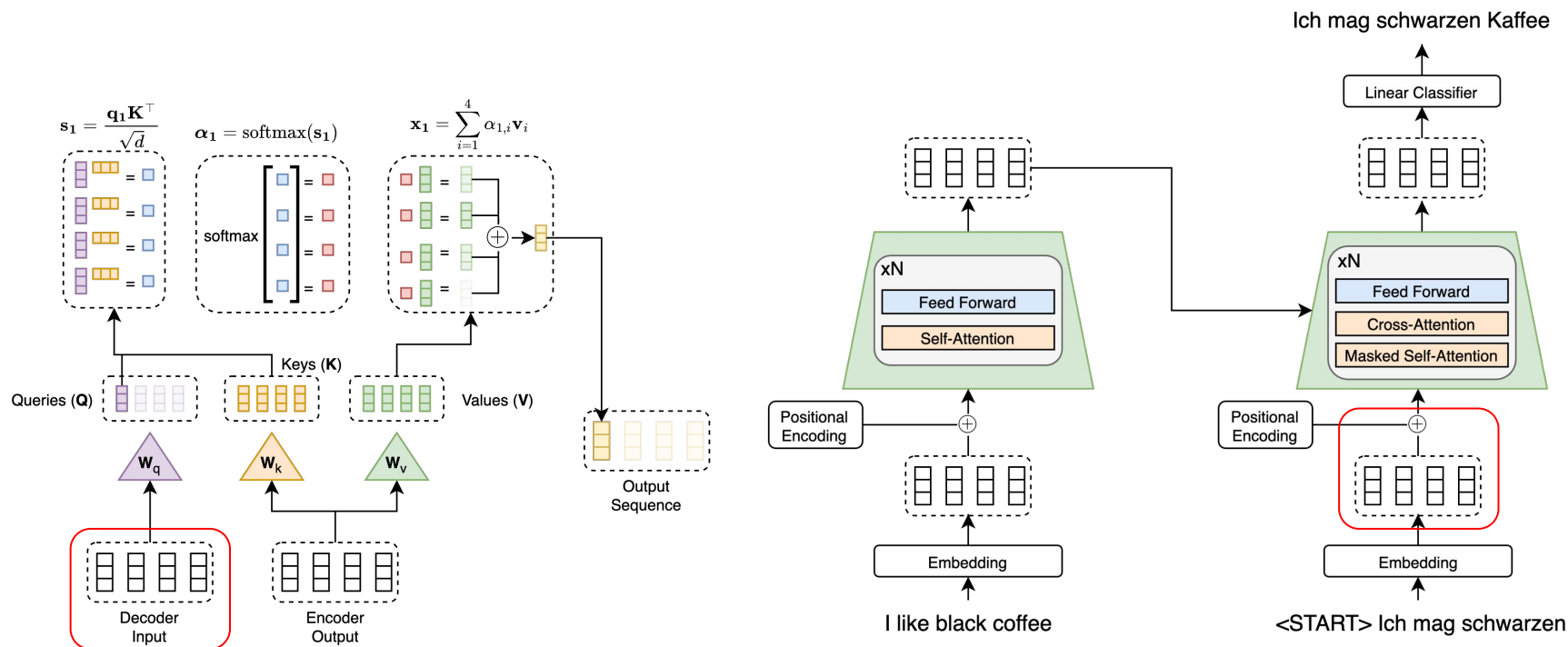
- Cannot be parallelized
- Potential solutions:
  - Speculative decoding
  - Combining AR models with Diffusion models
  - Spend more money



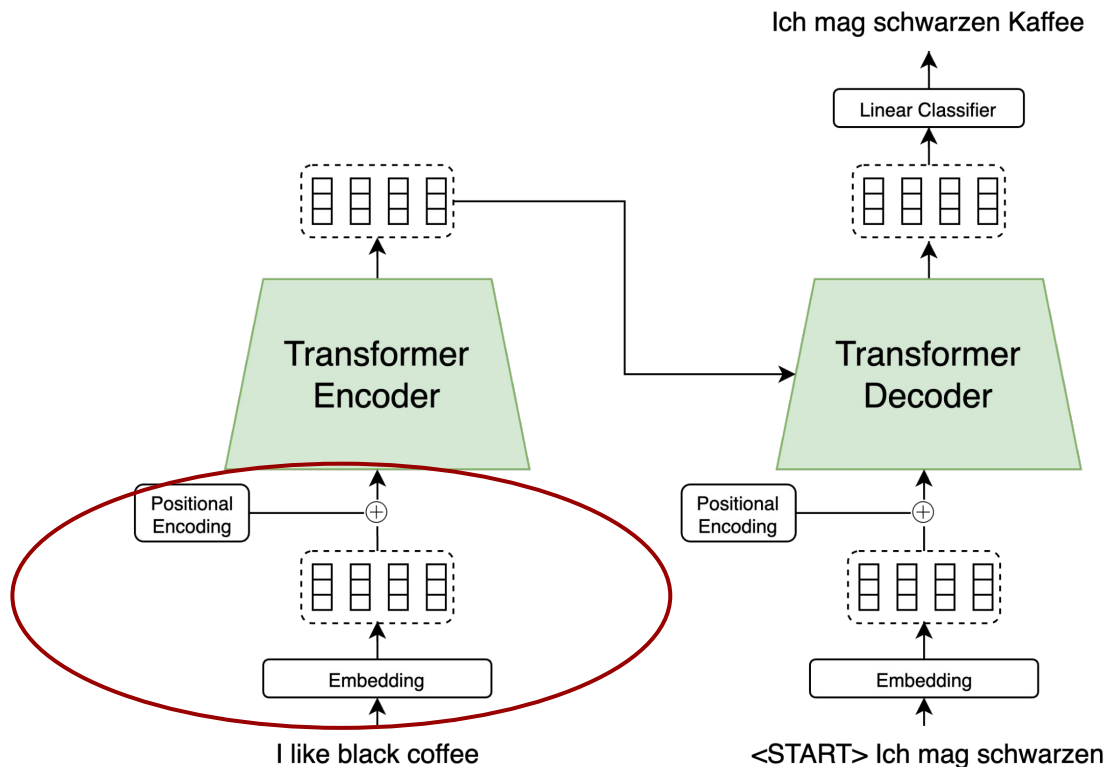
# Transformer Architecture



# Cross-Attention

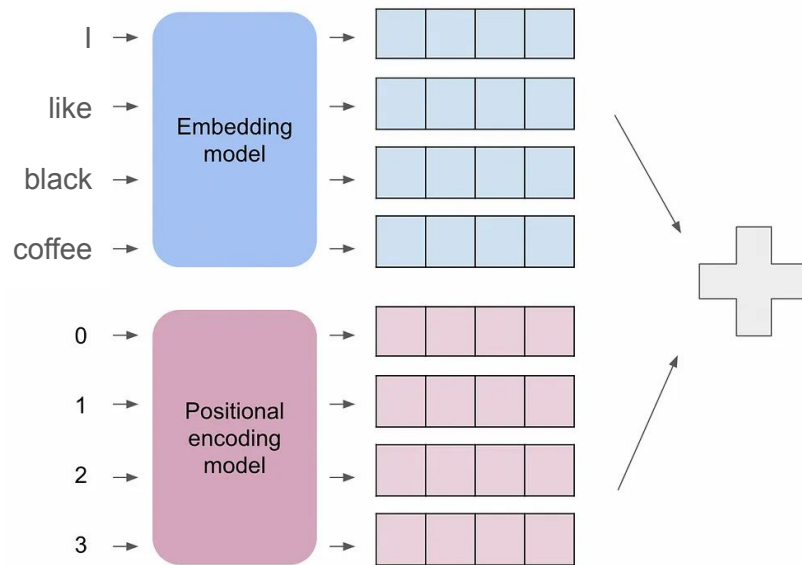


# Transformer Architecture



# Positional Encoding

- Embedding model:
  - Lookup table
  - Non-contextual word embeddings
- Problem:
  - Word order matters!
  - In attention word order is lost
- Solution:
  - Encode the **token position** as a vector
  - Add this vector to the **word embedding**



# Positional Encoding

Neural networks are not good with single dimensional values.  
Higher dimensions are better suited for hyper-plane based classifiers  
(internal to the neural network).

Naive Idea: Use binary encoding

| Token Position |   |   |   |   |   |   |   |   |
|----------------|---|---|---|---|---|---|---|---|
| Integer        | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| Binary         | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 |
|                | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 1 |
|                | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 |

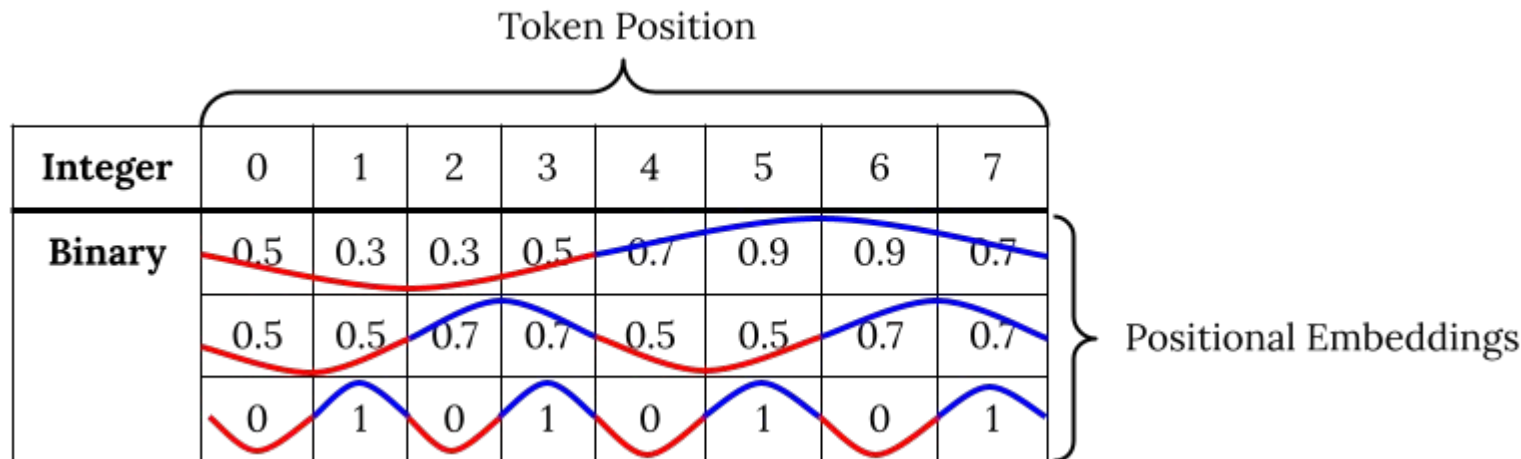
} Positional Embeddings

# Positional Encoding

Better idea: Use cosine / sine embeddings  
(Fourier features.)

For a position  $pos$  and embedding dimension  $i$ :

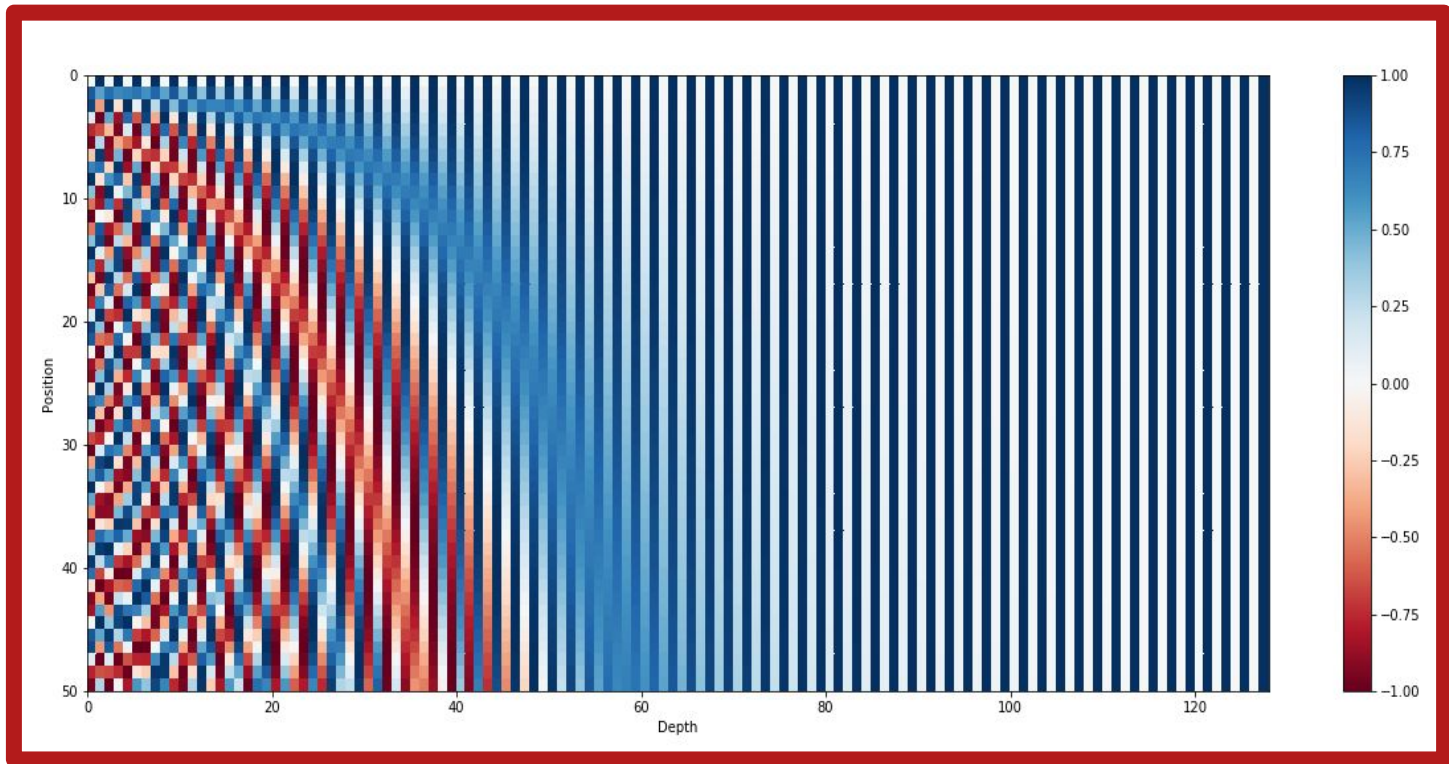
- $PE_{(pos, 2i)} = \sin\left(\frac{pos}{10000^{\frac{2i}{d_{model}}}}\right)$
- $PE_{(pos, 2i+1)} = \cos\left(\frac{pos}{10000^{\frac{2i}{d_{model}}}}\right)$



# Example Positional Encoding

For a position  $pos$  and embedding dimension  $i$ :

- $PE_{(pos, 2i)} = \sin\left(\frac{pos}{10000^{\frac{2i}{d_{\text{model}}}}}\right)$
- $PE_{(pos, 2i+1)} = \cos\left(\frac{pos}{10000^{\frac{2i}{d_{\text{model}}}}}\right)$



# Positional Encoding in Other Domains

---

## An intriguing failing of convolutional neural networks and the CoordConv solution

---

Rosanne Liu<sup>1</sup> Joel Lehman<sup>1</sup> Piero Molino<sup>1</sup> Felipe Petroski Such<sup>1</sup> Eric Frank<sup>1</sup>

Alex Sergeev<sup>2</sup>

Jason Yosinski<sup>1</sup>

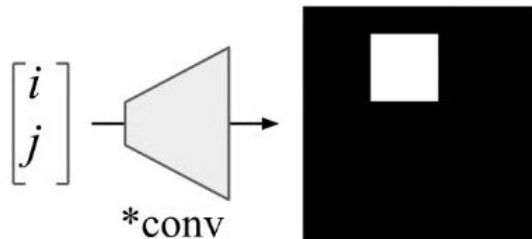
<sup>1</sup>Uber AI Labs, San Francisco, CA, USA    <sup>2</sup>Uber Technologies, Seattle, WA, USA

{rosanne, joel.lehman, piero, felipe.such, mysterefrank, asergeev, yosinski}@uber.com

### Supervised Rendering

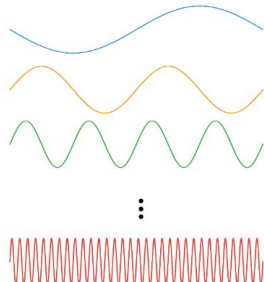
Output: per-pixel sigmoid

Loss: cross entropy

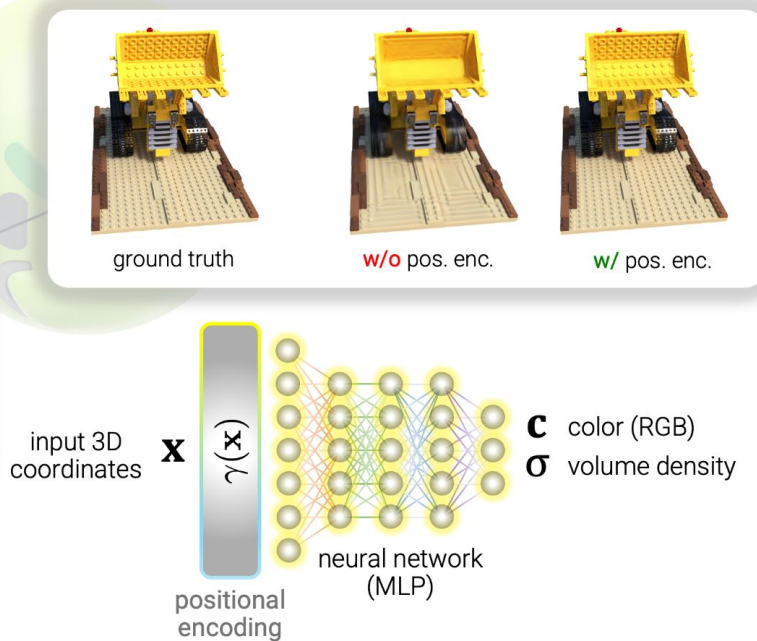


# Positional Encoding in Other Domains

## NeRF

$$\gamma(\mathbf{x}) = \begin{bmatrix} \gamma_0(\mathbf{x}) \\ \gamma_1(\mathbf{x}) \\ \vdots \\ \gamma_{L-1}(\mathbf{x}) \end{bmatrix}$$


where  $\gamma_k(\mathbf{x}) = [\cos(2^k \pi \mathbf{x}), \sin(2^k \pi \mathbf{x})]$

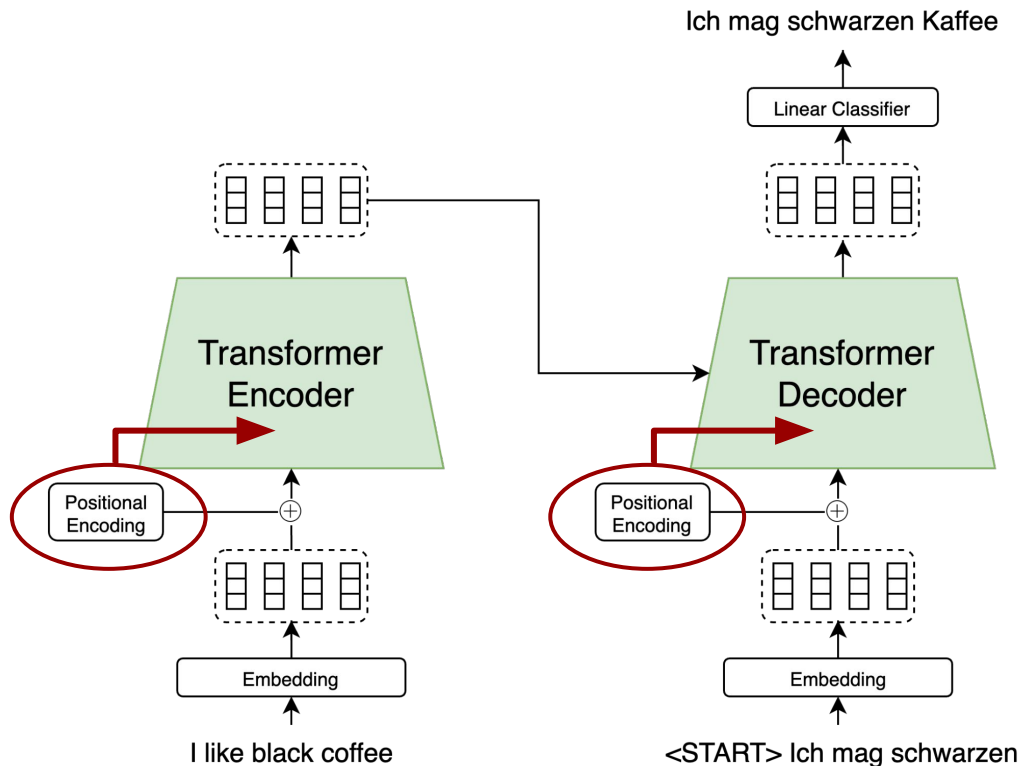


**Positional encoding** encourages representation of **higher-frequency signals**.

# Discuss!

- What are the potential issues of these type of position embedding?

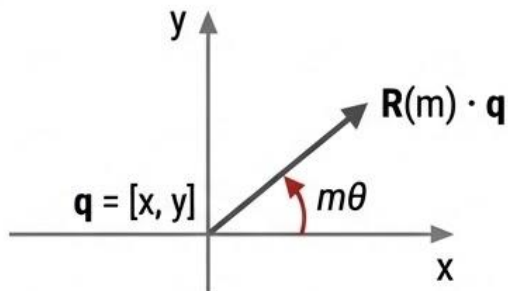
# Transformer Architecture





# RoPE: Rotary Position Embedding

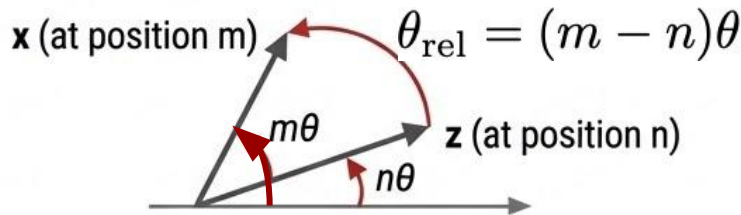
## The Core Rotation Mechanism



$$\mathbf{R}(m) \cdot \mathbf{q} = \begin{bmatrix} x \cdot \cos(m\theta) - y \cdot \sin(m\theta) \\ x \cdot \sin(m\theta) + y \cdot \cos(m\theta) \end{bmatrix}$$

A 2D rotation matrix  $\mathbf{R}(m)$  is applied to a query vector  $\mathbf{q}$ . The rotation angle  $m\theta$  is proportional to the position  $m$ .

## Preserving Relative Position

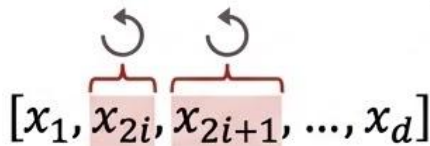


Rotation is **additive**:  $\mathbf{R}(m)\mathbf{R}(n)=\mathbf{R}(m+n)$  and  $\mathbf{R}(m)^T\mathbf{R}(n)=\mathbf{R}(m-n)$

$$\langle \mathbf{x}, \mathbf{z} \rangle = (\mathbf{R}(m) \cdot \mathbf{x})^T (\mathbf{R}(n) \cdot \mathbf{z}) = \mathbf{x}^T \mathbf{R}(m-n) \mathbf{z}$$

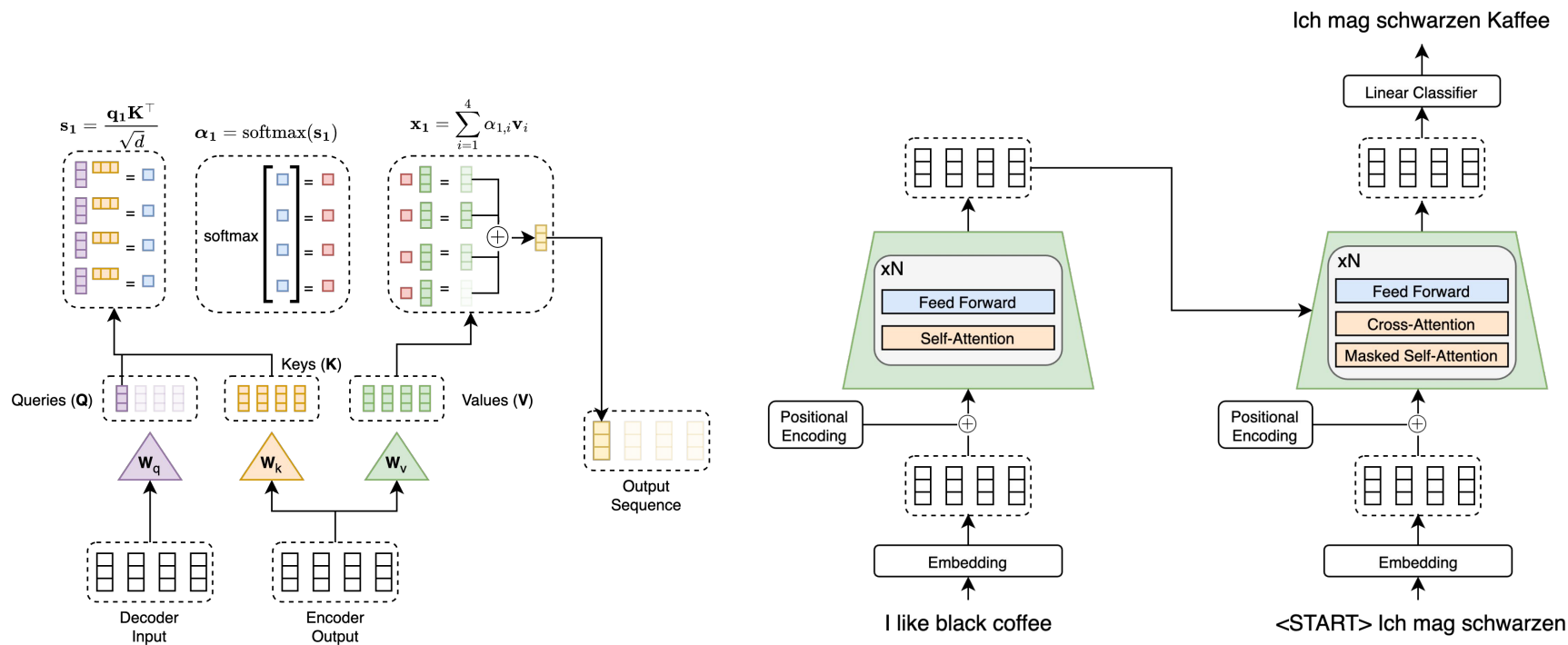
For vectors  $\mathbf{x}$  at position  $m$  and  $\mathbf{z}$  at position  $n$ , the dot product  $\langle \mathbf{x}, \mathbf{z} \rangle$  depends only on the relative distance  $(n-m)$ .

## Generalization to Higher Dimensions

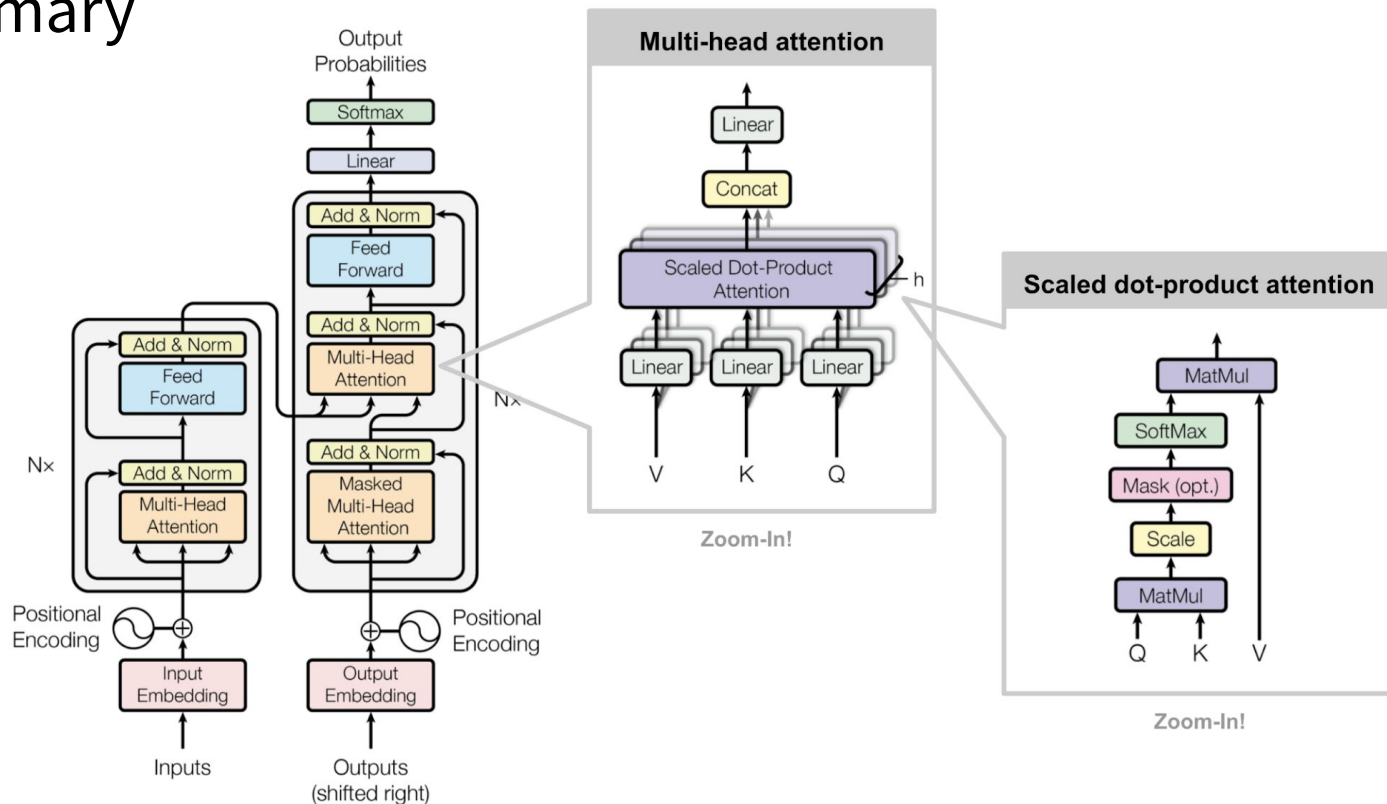


For higher dimensional vectors, apply the 2D rotation to pairs of dimensions. Define the angle of position  $i/2$  as  $\theta_i = \frac{1}{10000^{2i/d}}$ .

# Cross-Attention



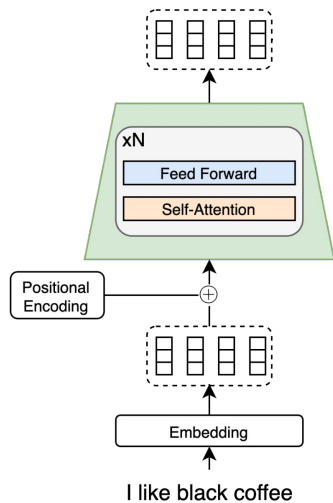
# Summary



# Three famous Transformer models

## BERT

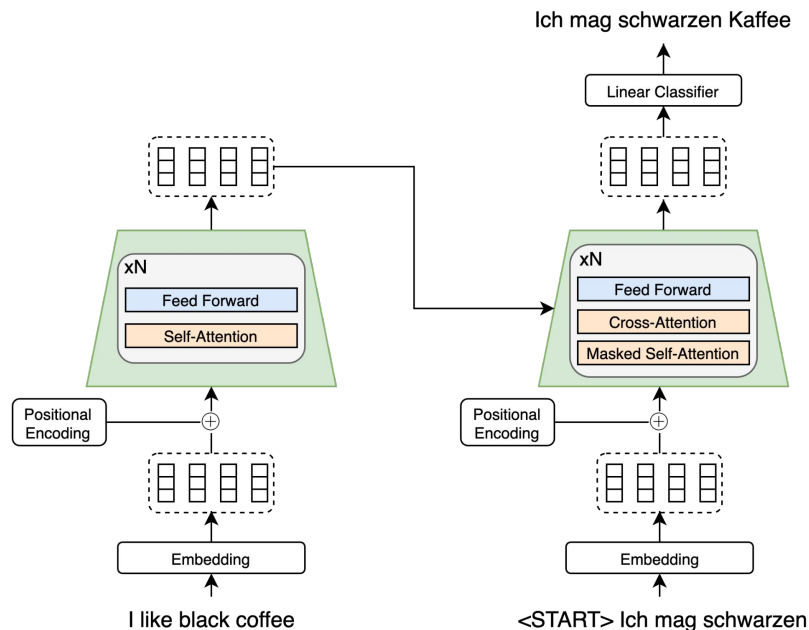
encoder only



[Devlin et al. 2019]

## T5/BART

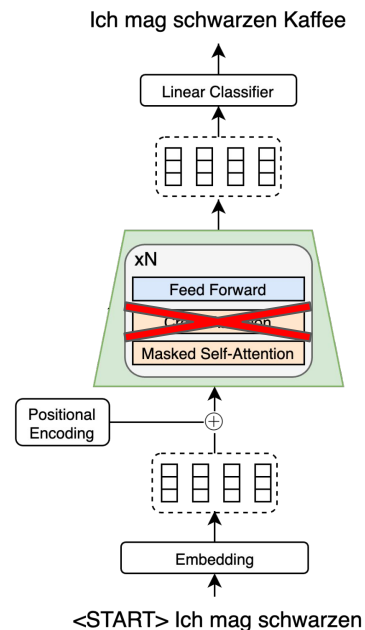
encoder-decoder model



[Raffel et al. 2020] [Lewis et al. 2020]

## GPT

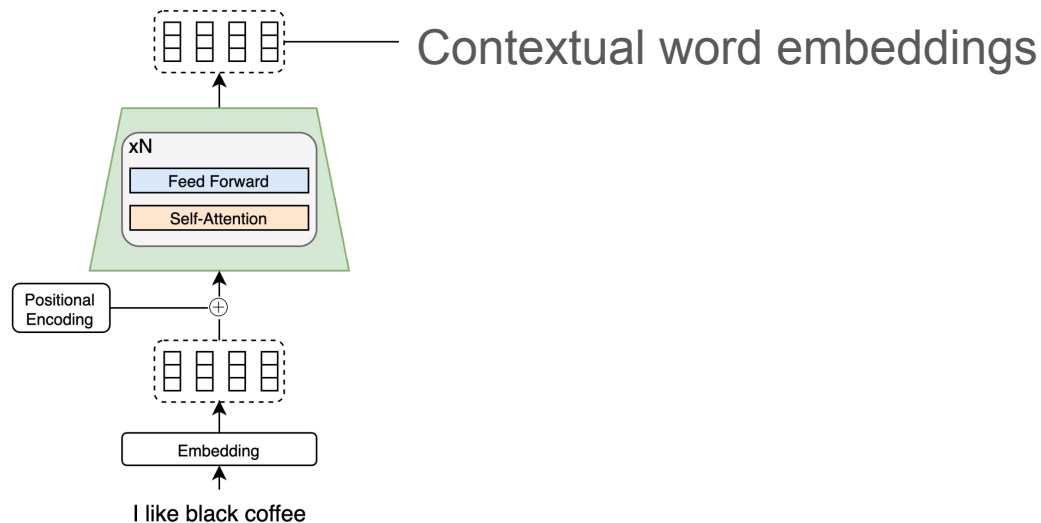
decoder only



[Radford et al. 2018]

# BERT (Bidirectional Encoder Representations from Transformers)

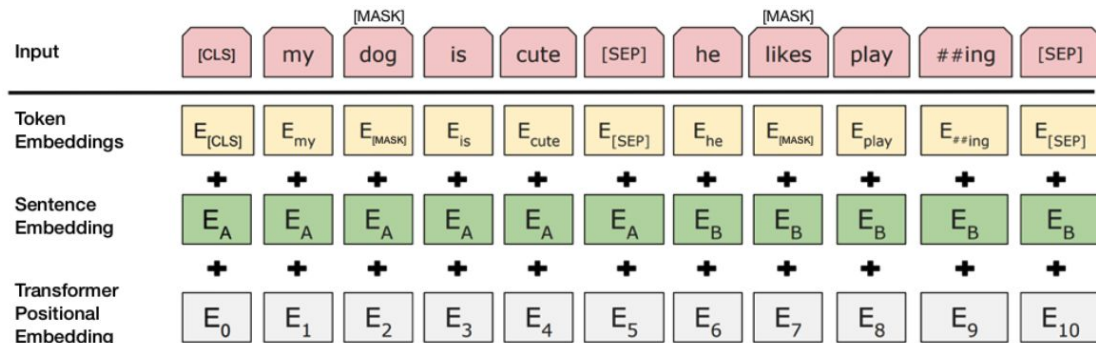
- Output of **Transformer Encoder** as contextual word embeddings
- Bidirectional Context
- Pre-trained on the language, and then fine-tuned



# BERT - Input Representation

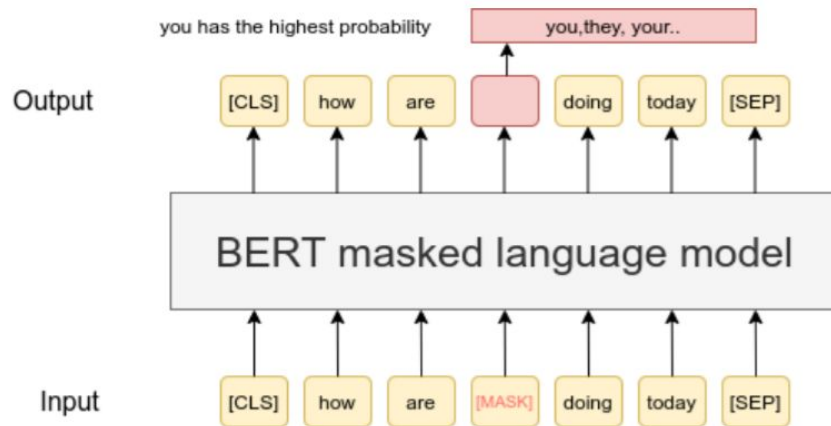
Input:

- Use 30,000 WordPiece vocabulary on input.
- Each token is sum of three embeddings



# Training

- Masked Language Modelling
  - Mask out k% of the input words, and then predict the masked words
  - the man went to the store to [MASK] a [MASK] of milk
  - What can you use as a loss function?



# Training

- Masked Language Modelling

- Mask out  $k\%$  of the input words, and then predict the masked words
- the man went to the store to [MASK] a [MASK] of milk
- What can you use as a loss function?

- Next sentence prediction

- To learn relationships between sentences, predict whether Sentence B is actual sentence that proceeds Sentence A, or a random sentence

**Sentence A** = The man went to the store.  
**Sentence B** = He bought a gallon of milk.  
**Label** = IsNextSentence

**Sentence A** = The man went to the store.  
**Sentence B** = Penguins are flightless.  
**Label** = NotNextSentence

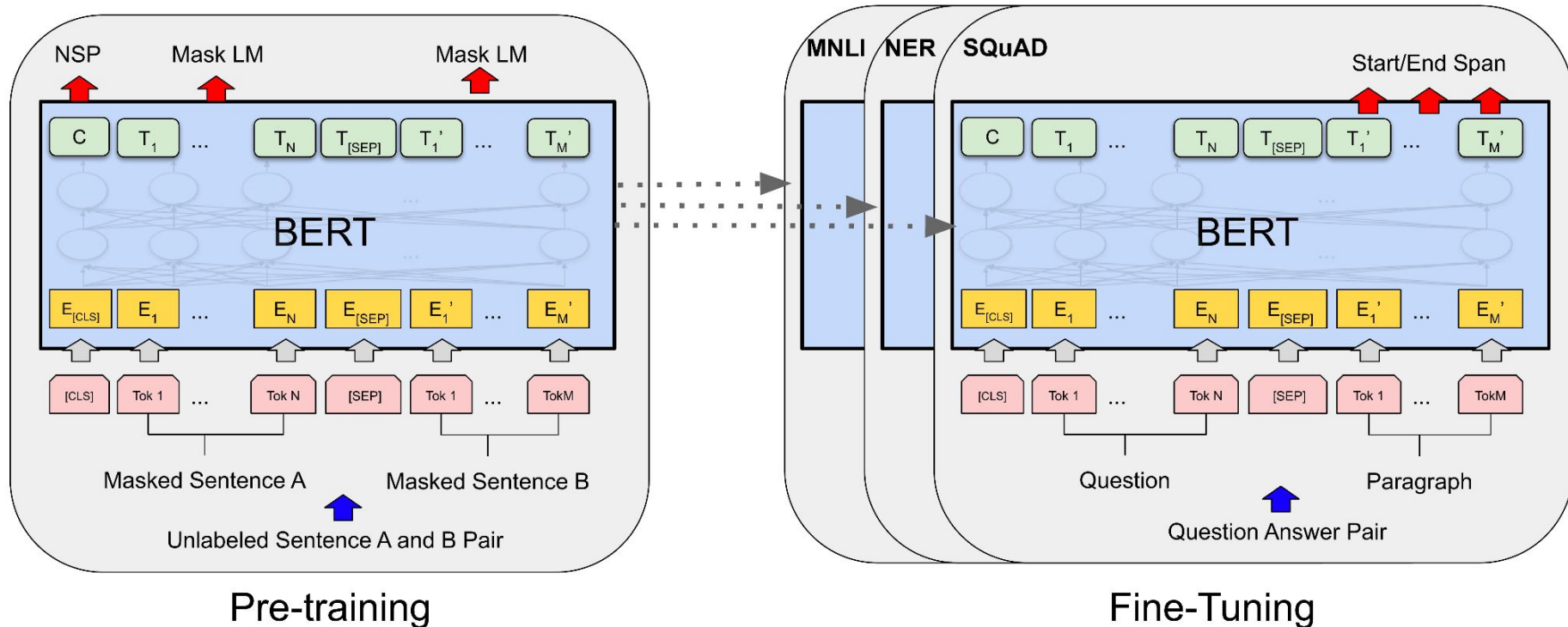
# Self-supervised Learning

- Labels are generated automatically, no human labeling process
- Benefits
  - Scales well
  - Cost-Efficient
  - Flexible
- Challenges
  - Larger datasets are required
  - More compute is necessary

## Model Details

- Data: Wikipedia (2.5B words) + BookCorpus (800M words)
- Batch Size: 131,072 words (1024 sequences \* 128 length or 256 sequences \* 512 length)
- Training Time: 1M steps (~40 epochs)
- Optimizer: AdamW,  $1e-4$  learning rate, linear decay
- BERT-Base: 12-layer, 768-hidden, 12-head, 110M params
- BERT-Large: 24-layer, 1024-hidden, 16-head, 340M params
- Trained on 4x4 or 8x8 TPU slice for 4 days

# Pre-training to Fine-tuning Pipeline



| System                | MNLI-(m/mm)<br>392k | QQP<br>363k | QNLI<br>108k | SST-2<br>67k | CoLA<br>8.5k | STS-B<br>5.7k | MRPC<br>3.5k | RTE<br>2.5k | Average<br>- |
|-----------------------|---------------------|-------------|--------------|--------------|--------------|---------------|--------------|-------------|--------------|
| Pre-OpenAI SOTA       | 80.6/80.1           | 66.1        | 82.3         | 93.2         | 35.0         | 81.0          | 86.0         | 61.7        | 74.0         |
| BiLSTM+ELMo+Attn      | 76.4/76.1           | 64.8        | 79.8         | 90.4         | 36.0         | 73.3          | 84.9         | 56.8        | 71.0         |
| OpenAI GPT            | 82.1/81.4           | 70.3        | 87.4         | 91.3         | 45.4         | 80.0          | 82.3         | 56.0        | 75.1         |
| BERT <sub>BASE</sub>  | 84.6/83.4           | 71.2        | 90.5         | 93.5         | 52.1         | 85.8          | 88.9         | 66.4        | 79.6         |
| BERT <sub>LARGE</sub> | <b>86.7/85.9</b>    | <b>72.1</b> | <b>92.7</b>  | <b>94.9</b>  | <b>60.5</b>  | <b>86.5</b>   | <b>89.3</b>  | <b>70.1</b> | <b>82.1</b>  |

# Pre-Train then Fine-tune Paradigm

- Pre-train a large model on a lot of data with some self-supervised objective.
  - E.g. Masked language modeling
- Fine-tune on (smaller) downstream datasets
- Benefits:
  - Transfer learning: Leveraging knowledge from self-supervised objective to improve performance
  - Avoids overfitting: Large models would overfit if trained from scratch

# Fine-tuning Limitations

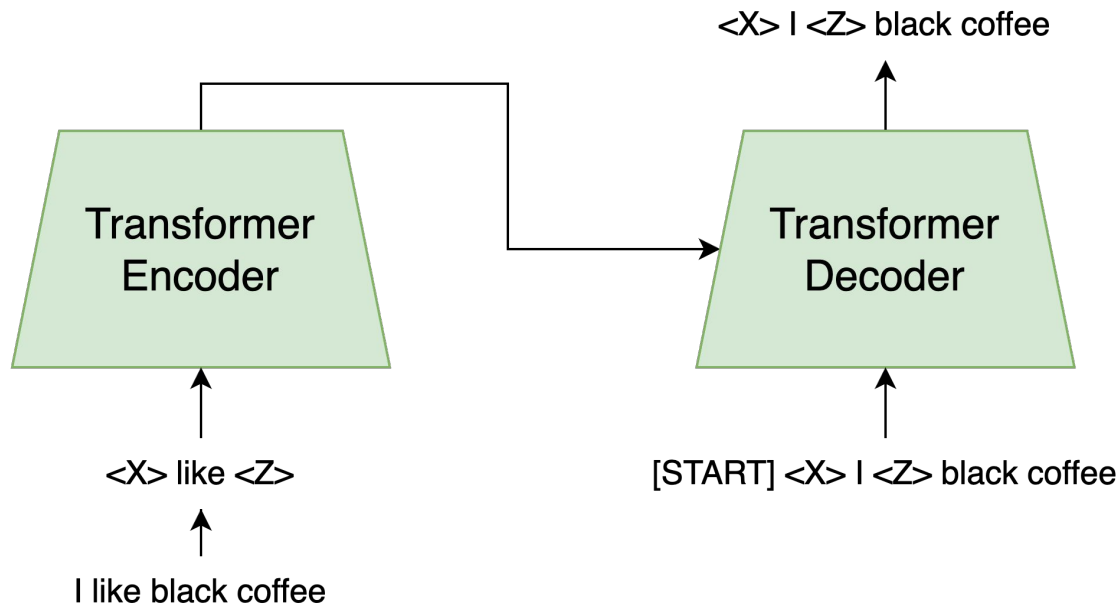
- Memory overhead
  - Training takes more memory than inference
  - Fine-tuning large models requires expensive GPUs
- Optimization
  - Might need to adjust optimization hyperparameters for each downstream task
  - Learning rates, batch sizes, etc.

## Question:

- What are some limitations of BERT?
  - Think about the different kinds of language tasks you might want to perform

# T5: Text-to-Text Transfer Transformer

- Pre-trained Encoder Decoder Language Model
  - Can generate text!



# T5 Span Corruption Objective

- Very similar to masked language modeling
- Key differences:
  - Mask spans with single “sentinel” tokens
  - Generate the masked text with a decoder

Original text

Thank you ~~for inviting~~ me to your party ~~last~~ week.

Inputs

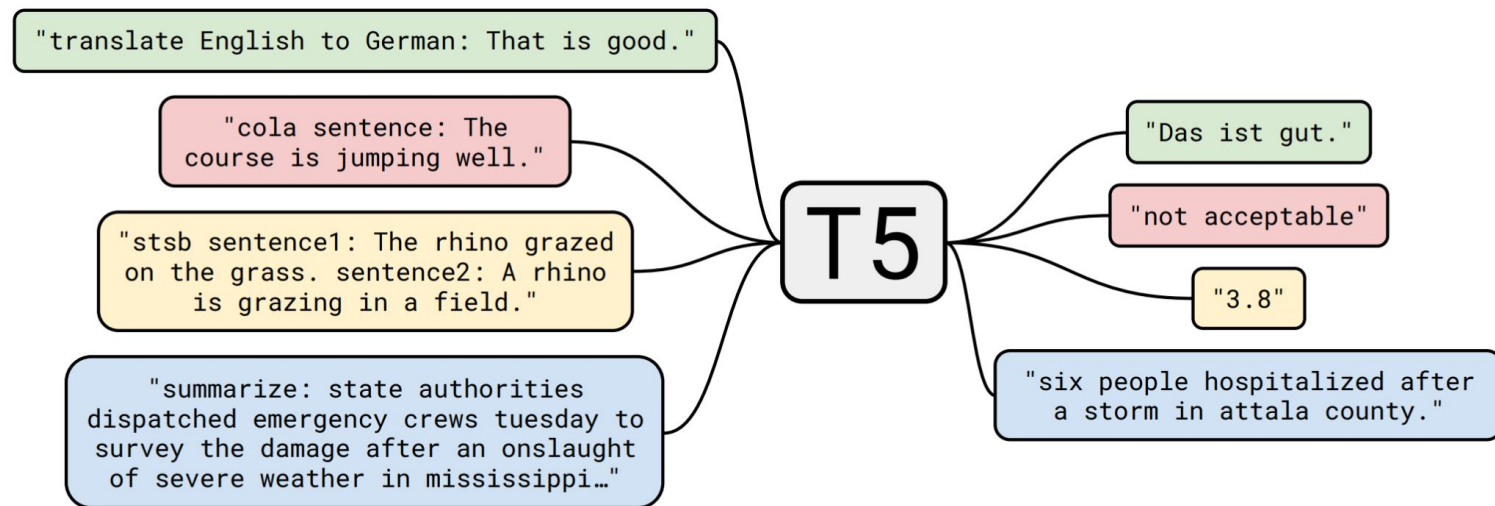
Thank you <X> me to your party <Y> week.

Targets

<X> for inviting <Y> last <Z>

# T5 Fine-Tuning

- Cast every task as a language generation problem
  - Even classification!



# Benchmark Results

- Outperformed best encoder-only models

| Model         | GLUE<br>Average   | CoLA<br>Matthew's | SST-2<br>Accuracy | MRPC<br>F1              | MRPC<br>Accuracy        | STS-B<br>Pearson  | STS-B<br>Spearman |
|---------------|-------------------|-------------------|-------------------|-------------------------|-------------------------|-------------------|-------------------|
| Previous best | 89.4 <sup>a</sup> | 69.2 <sup>b</sup> | 97.1 <sup>a</sup> | <b>93.6<sup>b</sup></b> | <b>91.5<sup>b</sup></b> | 92.7 <sup>b</sup> | 92.3 <sup>b</sup> |
| T5-Small      | 77.4              | 41.0              | 91.8              | 89.7                    | 86.6                    | 85.6              | 85.0              |
| T5-Base       | 82.7              | 51.1              | 95.2              | 90.7                    | 87.5                    | 89.4              | 88.6              |
| T5-Large      | 86.4              | 61.2              | 96.3              | 92.4                    | 89.9                    | 89.9              | 89.2              |
| T5-3B         | 88.5              | 67.1              | 97.4              | 92.5                    | 90.0                    | 90.6              | 89.8              |
| T5-11B        | <b>90.3</b>       | <b>71.6</b>       | <b>97.5</b>       | 92.8                    | 90.4                    | <b>93.1</b>       | <b>92.8</b>       |

| Model         | QQP<br>F1         | QQP<br>Accuracy         | MNLI-m<br>Accuracy | MNLI-mm<br>Accuracy | QNLI<br>Accuracy        | RTE<br>Accuracy   | WNLI<br>Accuracy  |
|---------------|-------------------|-------------------------|--------------------|---------------------|-------------------------|-------------------|-------------------|
| Previous best | 74.8 <sup>c</sup> | <b>90.7<sup>b</sup></b> | 91.3 <sup>a</sup>  | 91.0 <sup>a</sup>   | <b>99.2<sup>a</sup></b> | 89.2 <sup>a</sup> | 91.8 <sup>a</sup> |
| T5-Small      | 70.0              | 88.0                    | 82.4               | 82.3                | 90.3                    | 69.9              | 69.2              |
| T5-Base       | 72.6              | 89.4                    | 87.1               | 86.2                | 93.7                    | 80.1              | 78.8              |
| T5-Large      | 73.9              | 89.9                    | 89.9               | 89.6                | 94.8                    | 87.2              | 85.6              |
| T5-3B         | 74.4              | 89.7                    | 91.4               | 91.2                | 96.3                    | 91.1              | 89.7              |
| T5-11B        | <b>75.1</b>       | 90.6                    | <b>92.2</b>        | <b>91.9</b>         | 96.9                    | <b>92.8</b>       | <b>94.5</b>       |

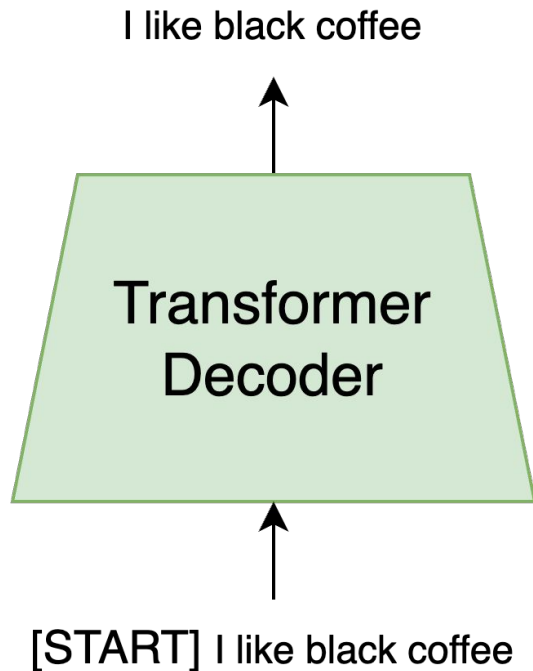
# T5 Ablations- Dataset Size

- More data helps!

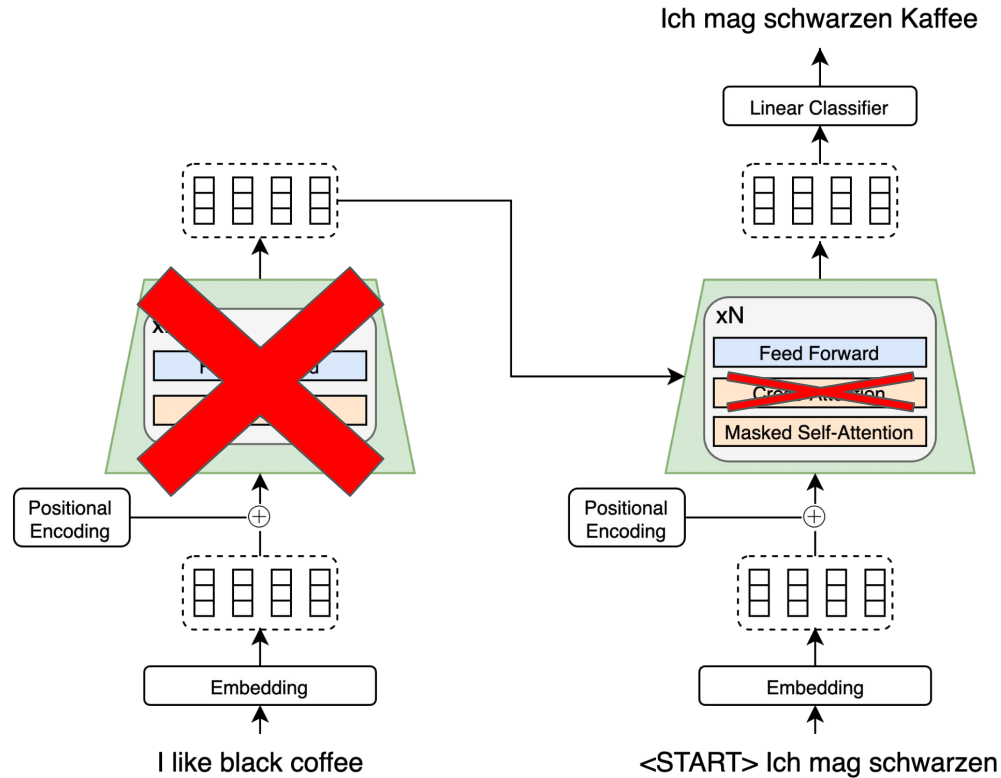
| Number of tokens | Repeats | GLUE         | CNNDM        | SQuAD        | SGLUE        | EnDe         | EnFr         | EnRo         |
|------------------|---------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
| ★ Full data set  | 0       | <b>83.28</b> | <b>19.24</b> | <b>80.88</b> | <b>71.36</b> | <b>26.98</b> | <b>39.82</b> | <b>27.65</b> |
| $2^{29}$         | 64      | <b>82.87</b> | <b>19.19</b> | <b>80.97</b> | <b>72.03</b> | <b>26.83</b> | <b>39.74</b> | <b>27.63</b> |
| $2^{27}$         | 256     | 82.62        | <b>19.20</b> | 79.78        | 69.97        | <b>27.02</b> | <b>39.71</b> | 27.33        |
| $2^{25}$         | 1,024   | 79.55        | 18.57        | 76.27        | 64.76        | 26.38        | 39.56        | 26.80        |
| $2^{23}$         | 4,096   | 76.34        | 18.33        | 70.92        | 59.29        | 26.37        | 38.84        | 25.81        |

# GPT

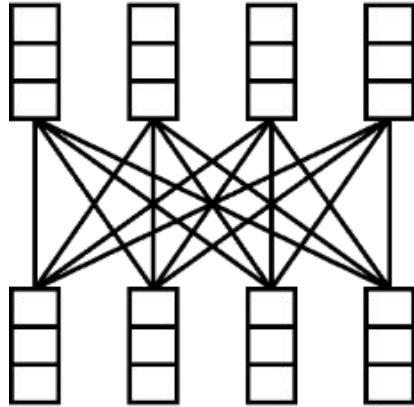
- Only use the transformer decoder
  - Train on next word prediction
  - “The capital of France is \_”
- Easy to apply in streaming settings
  - E.g. chatbots



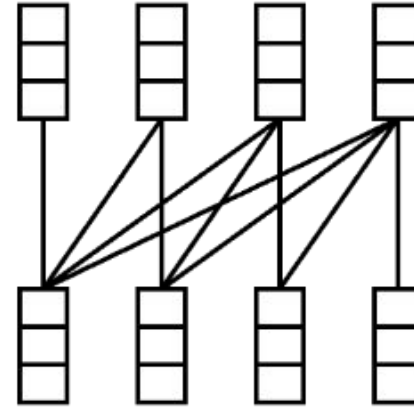
# Decoder-Only Architecture



## Review: Masked Self-Attention



Self-Attention



Masked Self-Attention

# GPT-2 Zero-Shot Capabilities

- Question-answering without any fine-tuning
  - Formatting is important!

## Context (passage and previous question/answer pairs)

Tom goes everywhere with Catherine Green, a 54-year-old secretary. He moves around her office at work and goes shopping with her. "Most people don't seem to mind Tom," says Catherine, who thinks he is wonderful. "He's my fourth child," she says. She may think of him and treat him that way as her son. He moves around buying his food, paying his health bills and his taxes, but in fact Tom is a dog.

Catherine and Tom live in Sweden, a country where everyone is expected to lead an orderly life according to rules laid down by the government, which also provides a high level of care for its people. This level of care costs money.

People in Sweden pay taxes on everything, so aren't surprised to find that owning a dog means more taxes. Some people are paying as much as 500 Swedish kronor in taxes a year for the right to keep their dog, which is spent by the government on dog hospitals and sometimes medical treatment for a dog that falls ill. However, most such treatment is expensive, so owners often decide to offer health and even life \_ for their dog.

In Sweden dog owners must pay for any damage their dog does. A Swedish Kennel Club official explains what this means: if your dog runs out on the road and gets hit by a passing car, you, as the owner, have to pay for any damage done to the car, even if your dog has been killed in the accident.

Q: How old is Catherine?

A: 54

Q: where does she live?

A:

**Model answer:** Stockholm

**Turker answers:** Sweden, Sweden, in Sweden, Sweden

# GPT-2

- Naturally occurring translation demonstrations in the pre-training corpus

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"I'm not the cleverest man in the world, but like they say in French: **Je ne suis pas un imbecile** [I'm not a fool].

In a now-deleted post from Aug. 16, Soheil Eid, Tory candidate in the riding of Joliette, wrote in French: "**Mentez mentez, il en restera toujours quelque chose**," which translates as, "**Lie lie and something will always remain.**"

"I hate the word '**perfume**,'" Burr says. 'It's somewhat better in French: '**parfum.**'

If listened carefully at 29:55, a conversation can be heard between two guys in French: "**-Comment on fait pour aller de l'autre côté? -Quel autre côté?**", which means "**- How do you get to the other side? - What side?**".

If this sounds like a bit of a stretch, consider this question in French: **As-tu aller au cinéma?**, or **Did you go to the movies?**, which literally translates as Have-you to go to movies/theater?

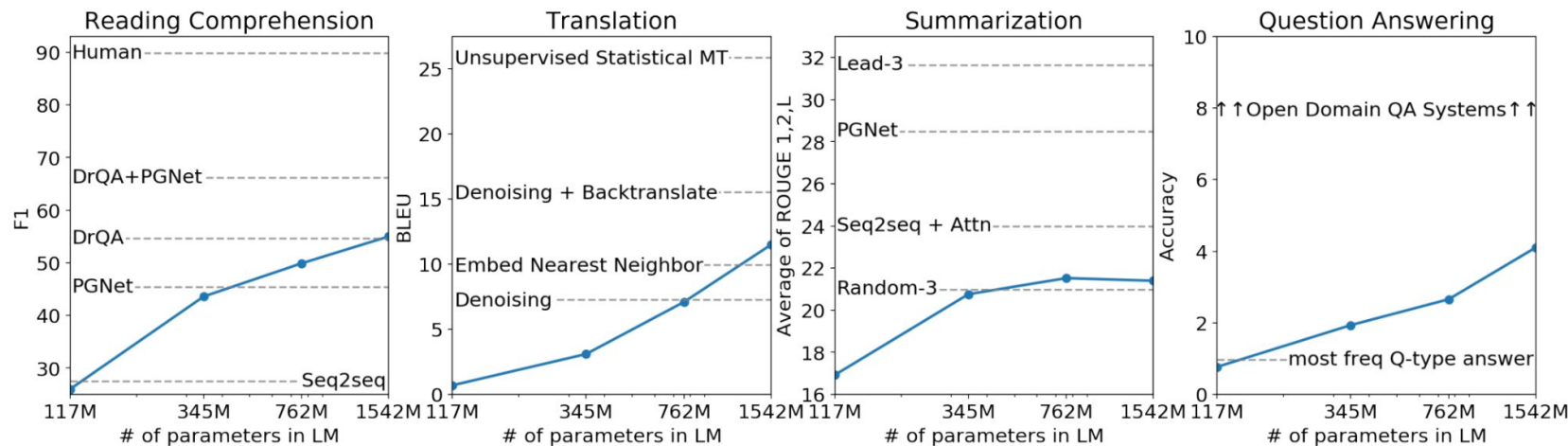
**"Brevet Sans Garantie Du Gouvernement"**, translated to English: **"Patented without government warranty"**.

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*Table 1.* Examples of naturally occurring demonstrations of English to French and French to English translation found throughout the WebText training set.

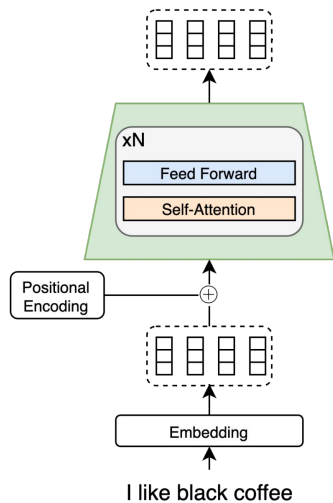
# GPT-2

- Train self-supervised auto-regressive LMs on web text
  - Next-word prediction
  - Up to 1.5 billion parameters
- Observed non-trivial zero-shot performance



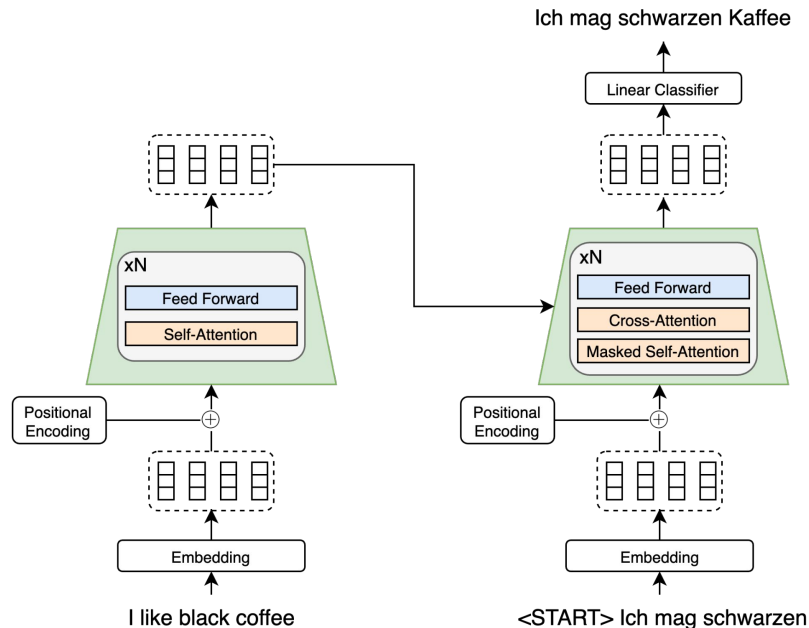
## BERT

encoder only



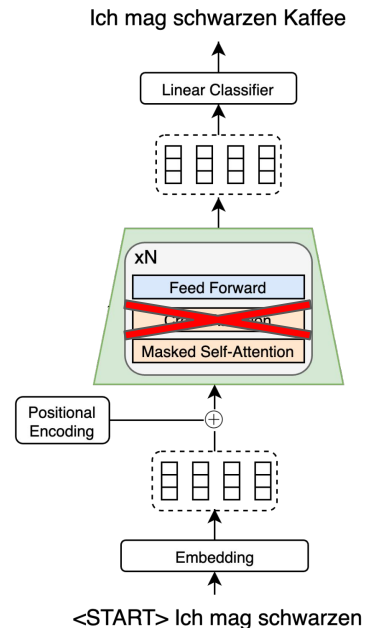
## T5/BART

encoder-decoder model



## GPT

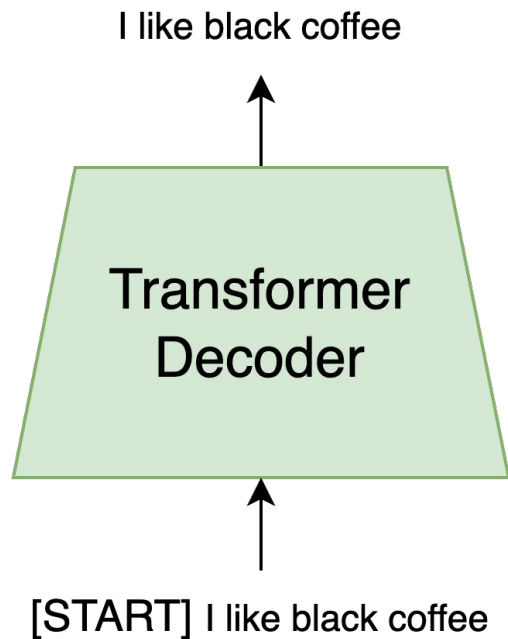
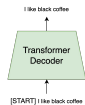
decoder only



# Scaling up!!

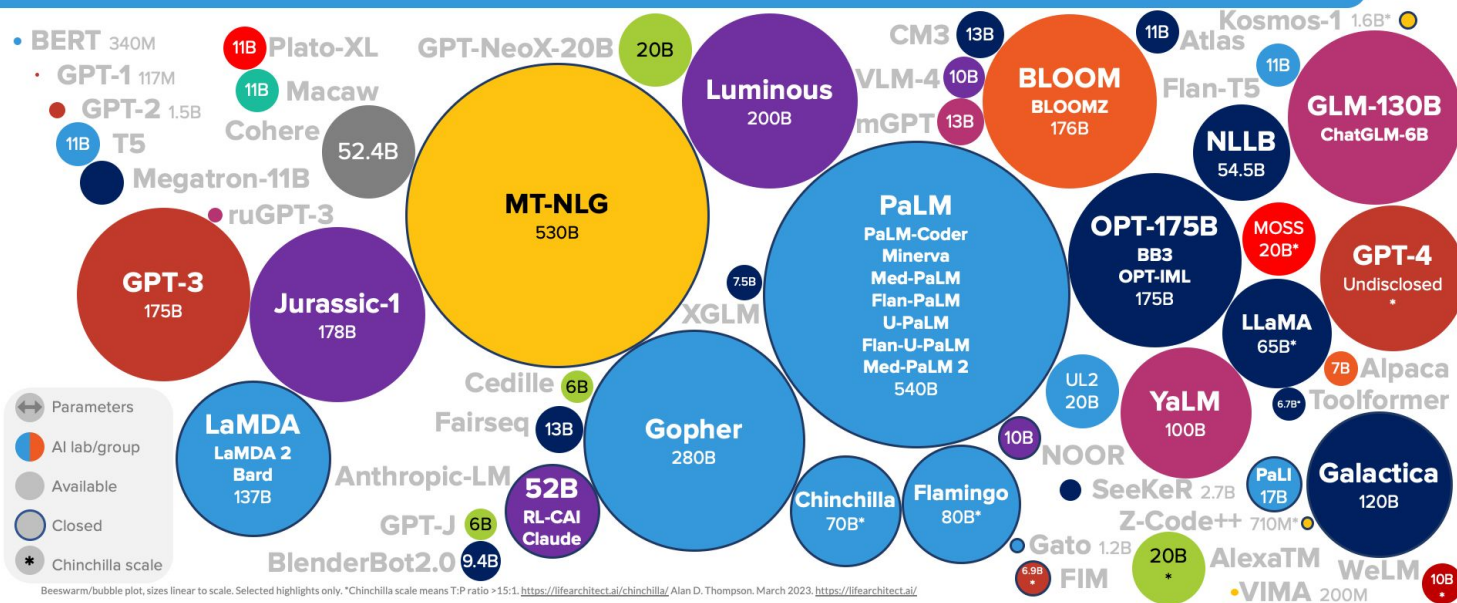
# GPT-3

- Train a bigger version of GPT-2 on more data



## GPT-3

## LANGUAGE MODEL SIZES TO MAR/2023

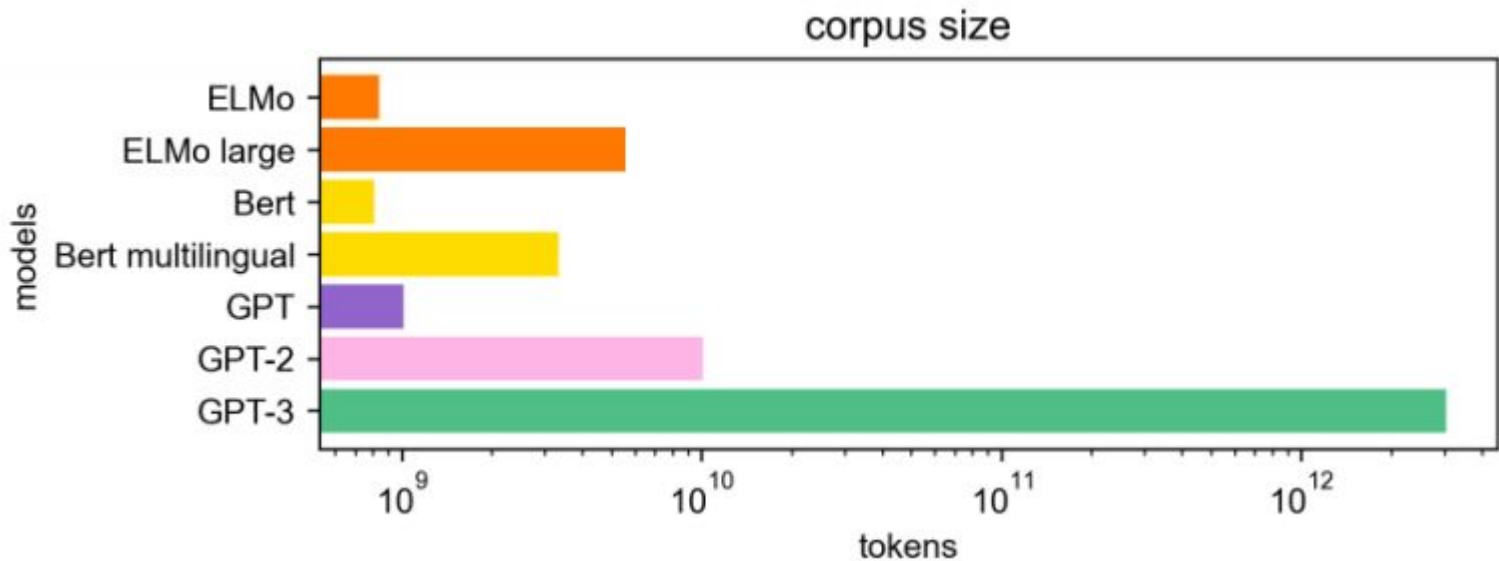


## GPT-3

| Model Name            | $n_{\text{params}}$ | $n_{\text{layers}}$ | $d_{\text{model}}$ | $n_{\text{heads}}$ | $d_{\text{head}}$ | Batch Size | Learning Rate        |
|-----------------------|---------------------|---------------------|--------------------|--------------------|-------------------|------------|----------------------|
| GPT-3 Small           | 125M                | 12                  | 768                | 12                 | 64                | 0.5M       | $6.0 \times 10^{-4}$ |
| GPT-3 Medium          | 350M                | 24                  | 1024               | 16                 | 64                | 0.5M       | $3.0 \times 10^{-4}$ |
| GPT-3 Large           | 760M                | 24                  | 1536               | 16                 | 96                | 0.5M       | $2.5 \times 10^{-4}$ |
| GPT-3 XL              | 1.3B                | 24                  | 2048               | 24                 | 128               | 1M         | $2.0 \times 10^{-4}$ |
| GPT-3 2.7B            | 2.7B                | 32                  | 2560               | 32                 | 80                | 1M         | $1.6 \times 10^{-4}$ |
| GPT-3 6.7B            | 6.7B                | 32                  | 4096               | 32                 | 128               | 2M         | $1.2 \times 10^{-4}$ |
| GPT-3 13B             | 13.0B               | 40                  | 5140               | 40                 | 128               | 2M         | $1.0 \times 10^{-4}$ |
| GPT-3 175B or “GPT-3” | 175.0B              | 96                  | 12288              | 96                 | 128               | 3.2M       | $0.6 \times 10^{-4}$ |

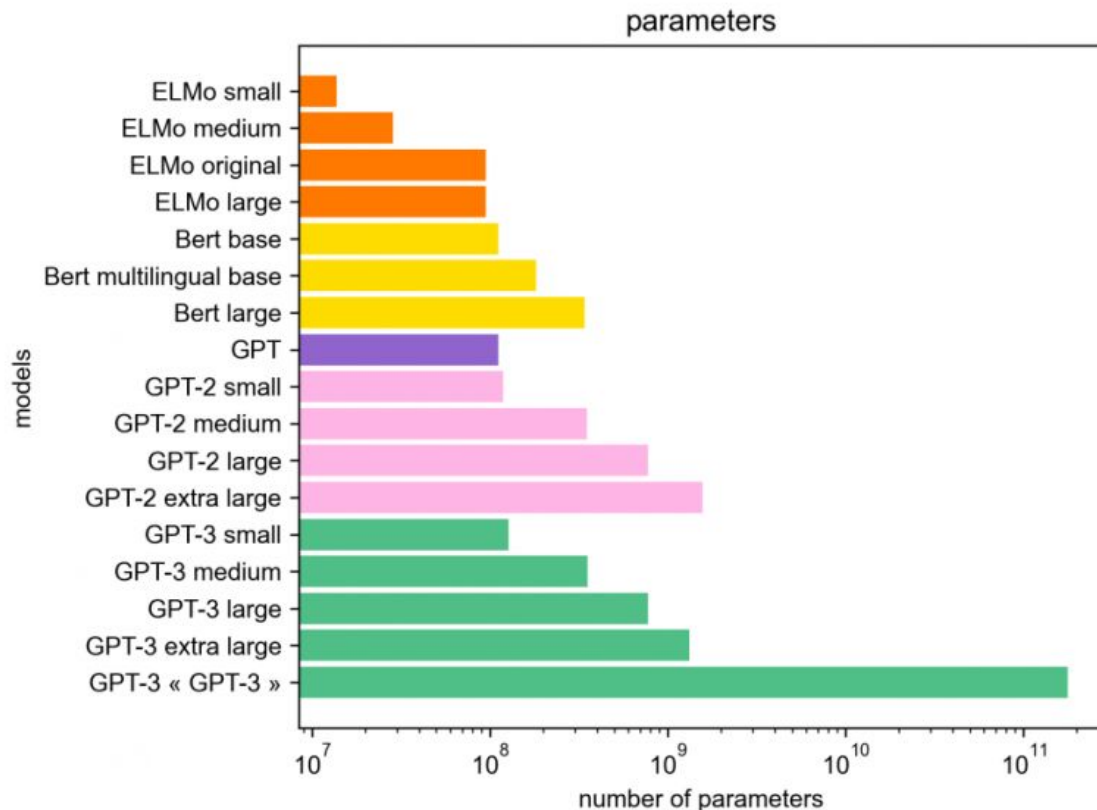
**Table 2.1:** Sizes, architectures, and learning hyper-parameters (batch size in tokens and learning rate) of the models which we trained. All models were trained for a total of 300 billion tokens.

## Scaling Training **Data**



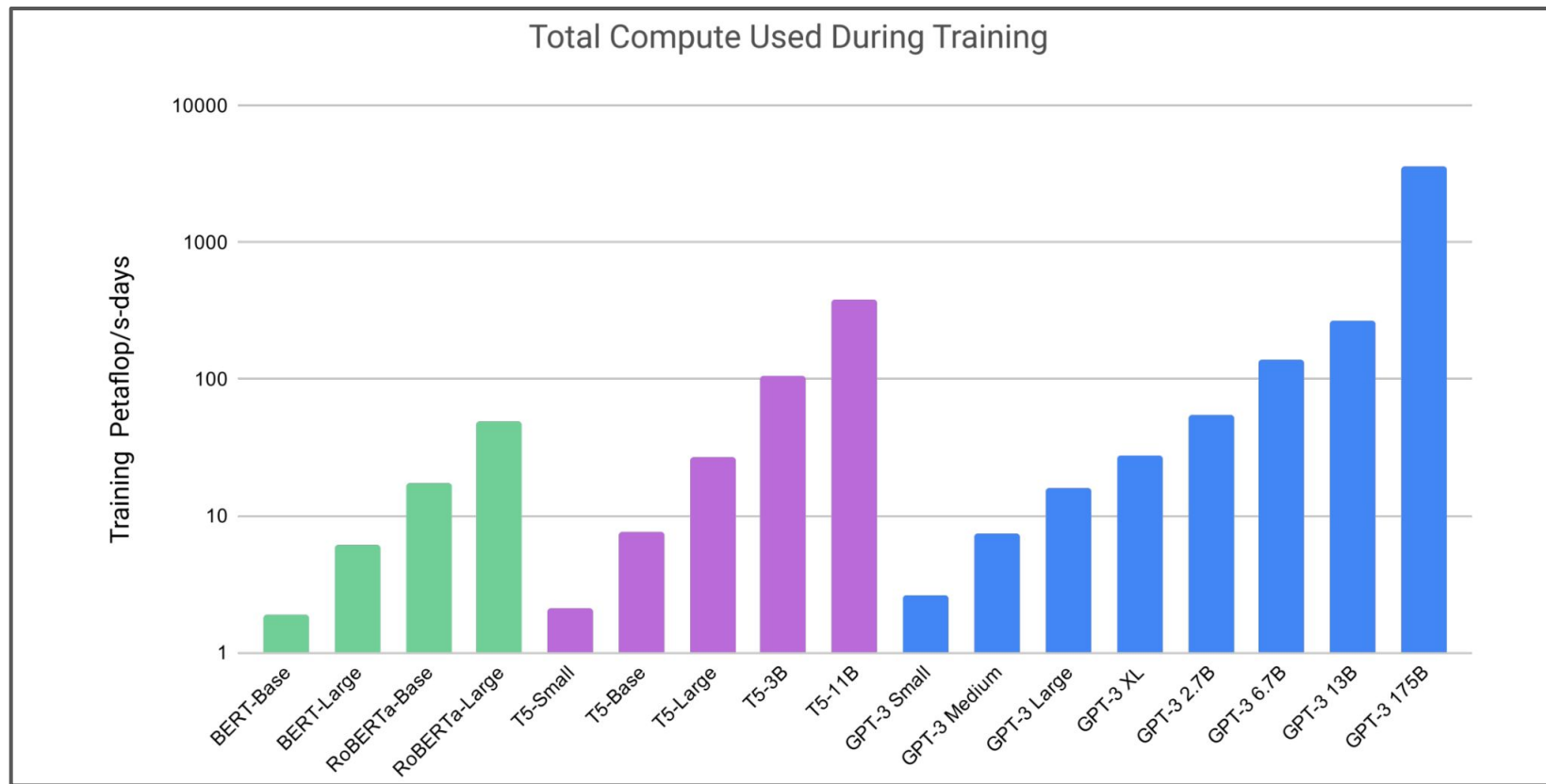
Source: <https://hellofuture.orange.com/en/the-gpt-3-language-model-revolution-or-evolution/>

# Scaling **Model** Size



Source: <https://hellofuture.orange.com/en/the-gpt-3-language-model-revolution-or-evolution/>

# Scaling Training Compute



# Discuss

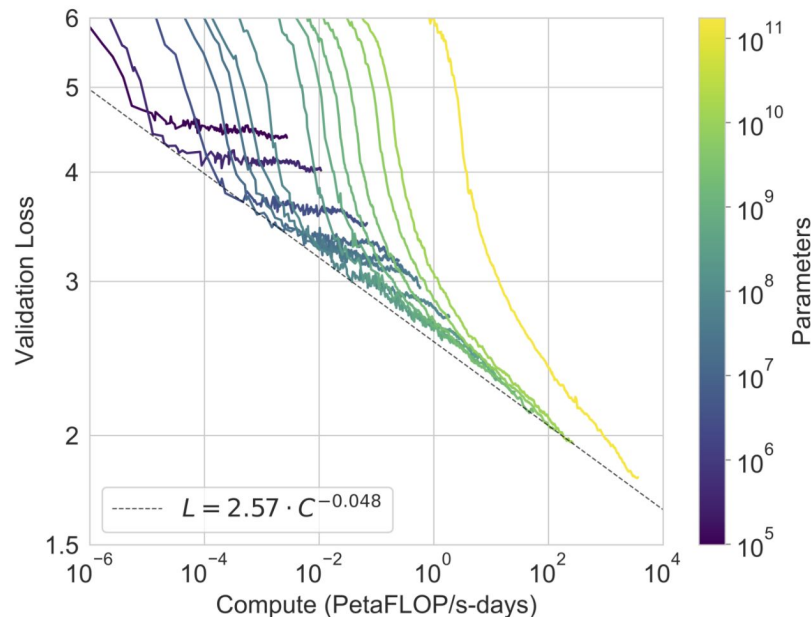
- How many parameters in a single feedforward layer of GPT-3
  - An MLP with:
    - Input dimension of 12,228
    - One hidden layer with dimension  $4 \times 12,228$
    - Output dimension of 12,228

| Model Name            | $n_{\text{params}}$ | $n_{\text{layers}}$ | $d_{\text{model}}$ | $n_{\text{heads}}$ | $d_{\text{head}}$ | Batch Size | Learning Rate        |
|-----------------------|---------------------|---------------------|--------------------|--------------------|-------------------|------------|----------------------|
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**Table 2.1:** Sizes, architectures, and learning hyper-parameters (batch size in tokens and learning rate) of the models which we trained. All models were trained for a total of 300 billion tokens.

# Effect of Scaling on Validation Loss

[J. Kaplan et al. (2020)]



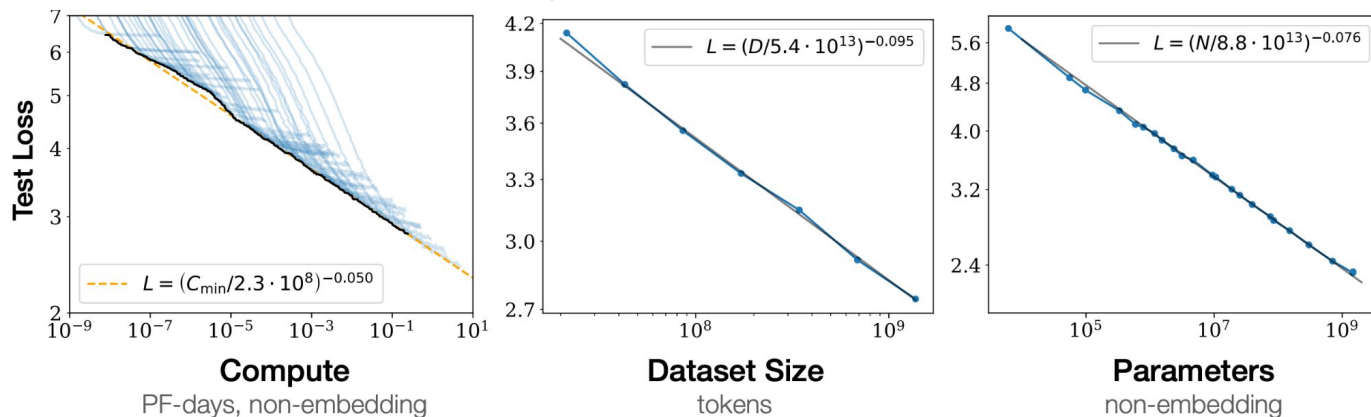
**Figure 3.1: Smooth scaling of performance with compute.** Performance (measured in terms of cross-entropy validation loss) follows a power-law trend with the amount of compute used for training. The power-law behavior observed in [KMH<sup>+</sup>20] continues for an additional two orders of magnitude with only small deviations from the predicted curve. For this figure, we exclude embedding parameters from compute and parameter counts.

# Scaling Laws

$$C = C_0 N D$$

$$L = \frac{A}{N^\alpha} + \frac{B}{D^\beta} + L_0$$

- Performance improves predictably with increased compute, data, and parameters
  - Can actually fit power laws!
  - Predict performance before training!



**Figure 1** Language modeling performance improves smoothly as we increase the model size, dataset size, and amount of compute<sup>2</sup> used for training. For optimal performance all three factors must be scaled up in tandem. Empirical performance has a power-law relationship with each individual factor when not bottlenecked by the other two.

# GPT-3 Evaluation

## Zero-shot

The model predicts the answer given only a natural language description of the task. No gradient updates are performed.

```
1 Translate English to French: ← task description
2 cheese => ..... ← prompt
```

## One-shot

In addition to the task description, the model sees a single example of the task. No gradient updates are performed.

```
1 Translate English to French: ← task description
2 sea otter => loutre de mer ← example
3 cheese => ..... ← prompt
```

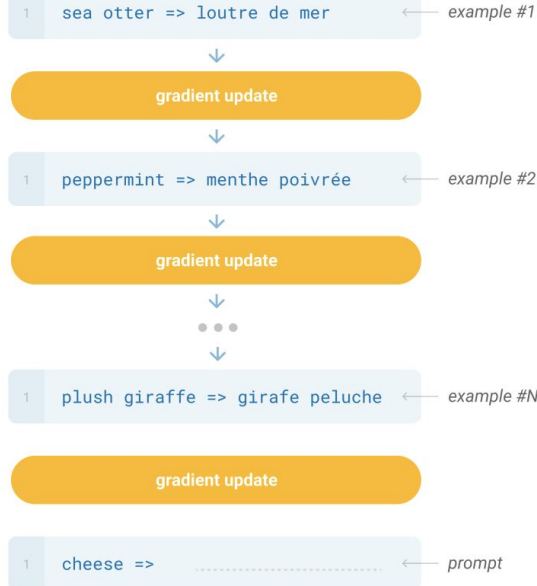
## Few-shot

In addition to the task description, the model sees a few examples of the task. No gradient updates are performed.

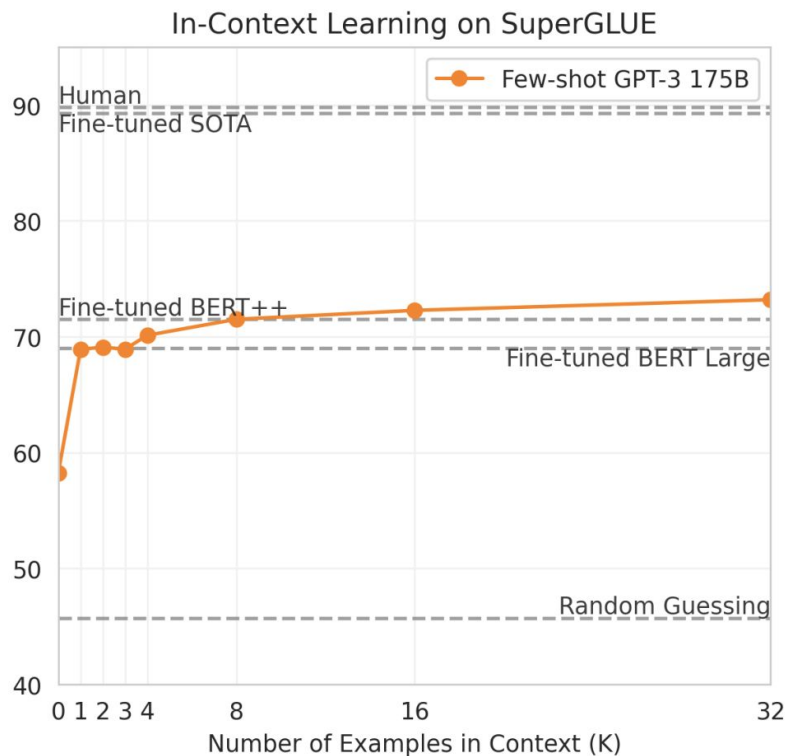
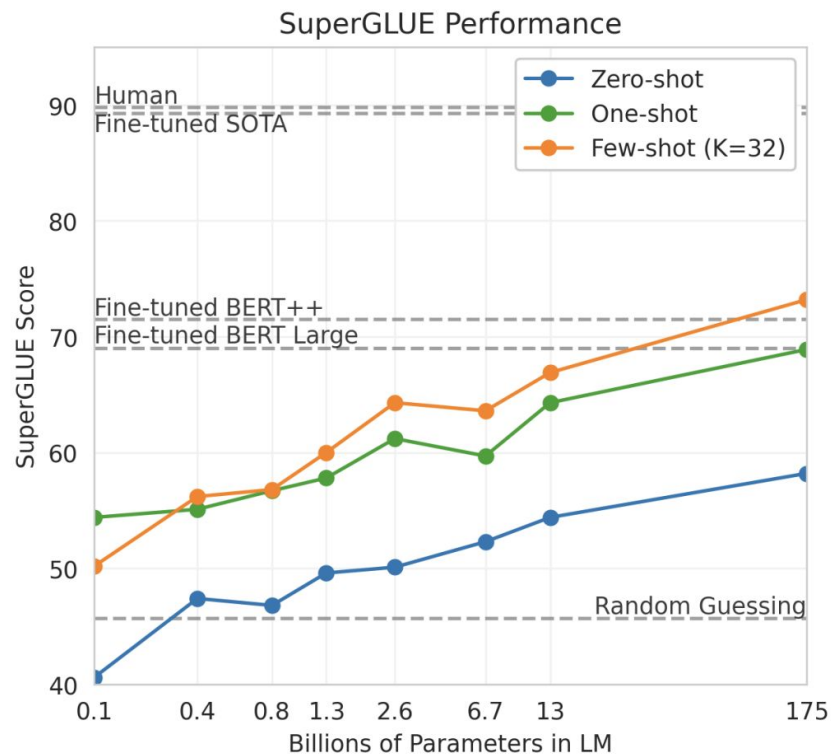
```
1 Translate English to French: ← task description
2 sea otter => loutre de mer ← examples
3 peppermint => menthe poivrée ←
4 plush girafe => girafe peluche ←
5 cheese => ..... ← prompt
```

## Fine-tuning

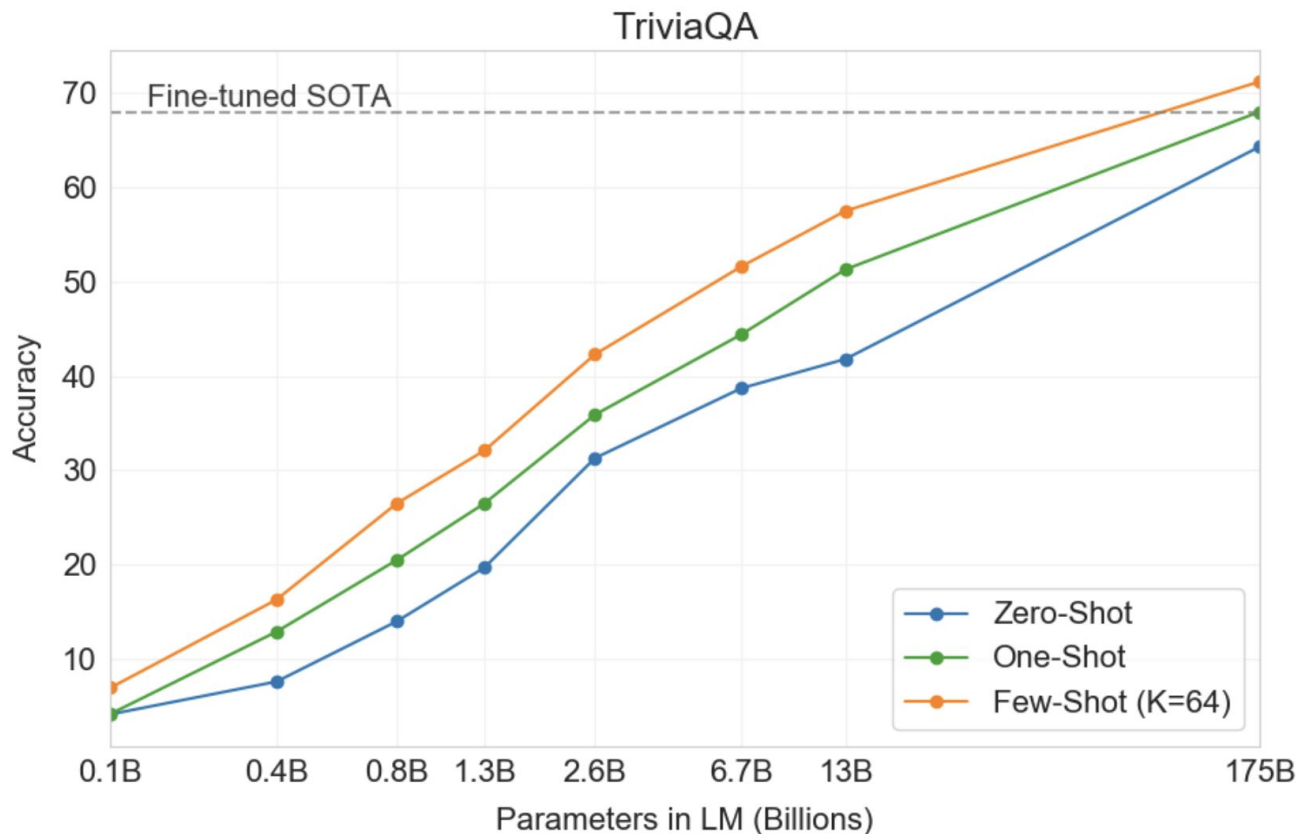
The model is trained via repeated gradient updates using a large corpus of example tasks.



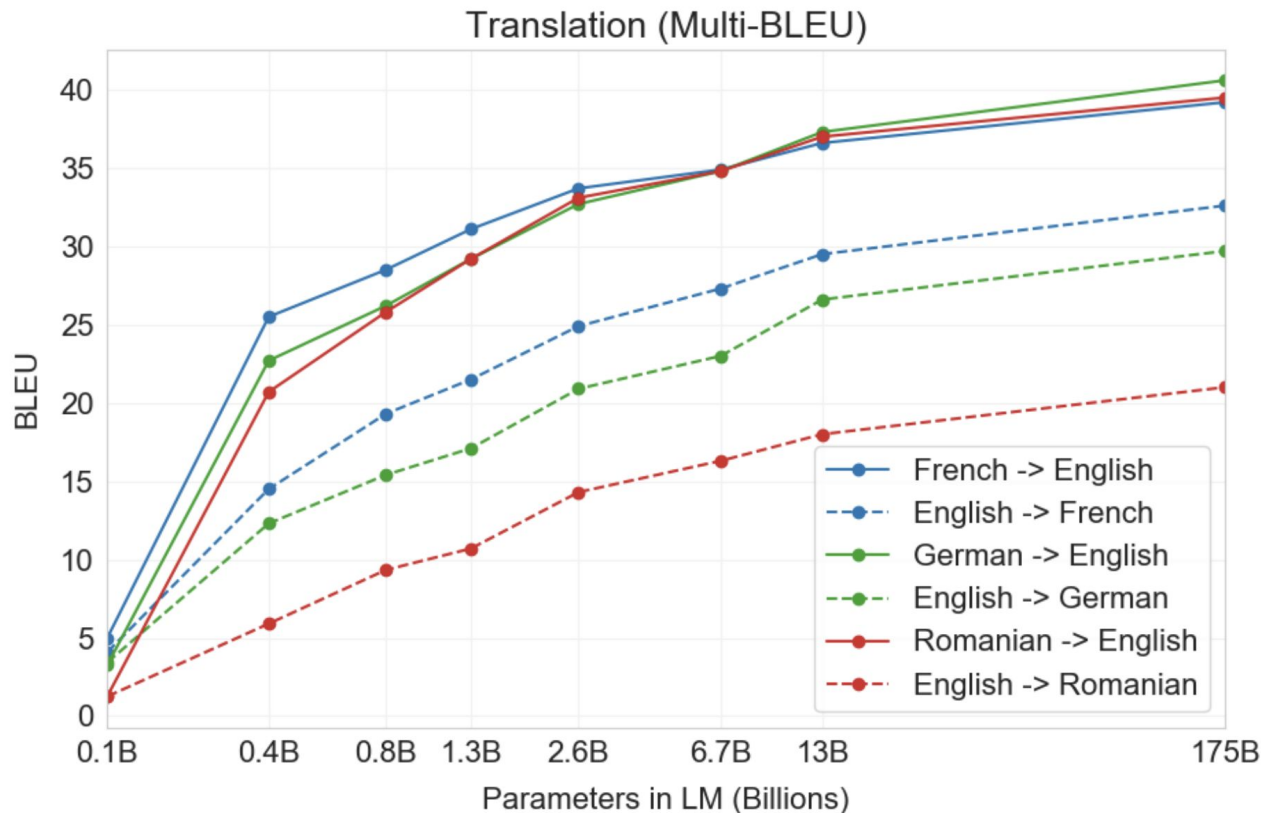
# GPT-3 Evaluation



# GPT-3 Evaluation



# GPT-3 Evaluation



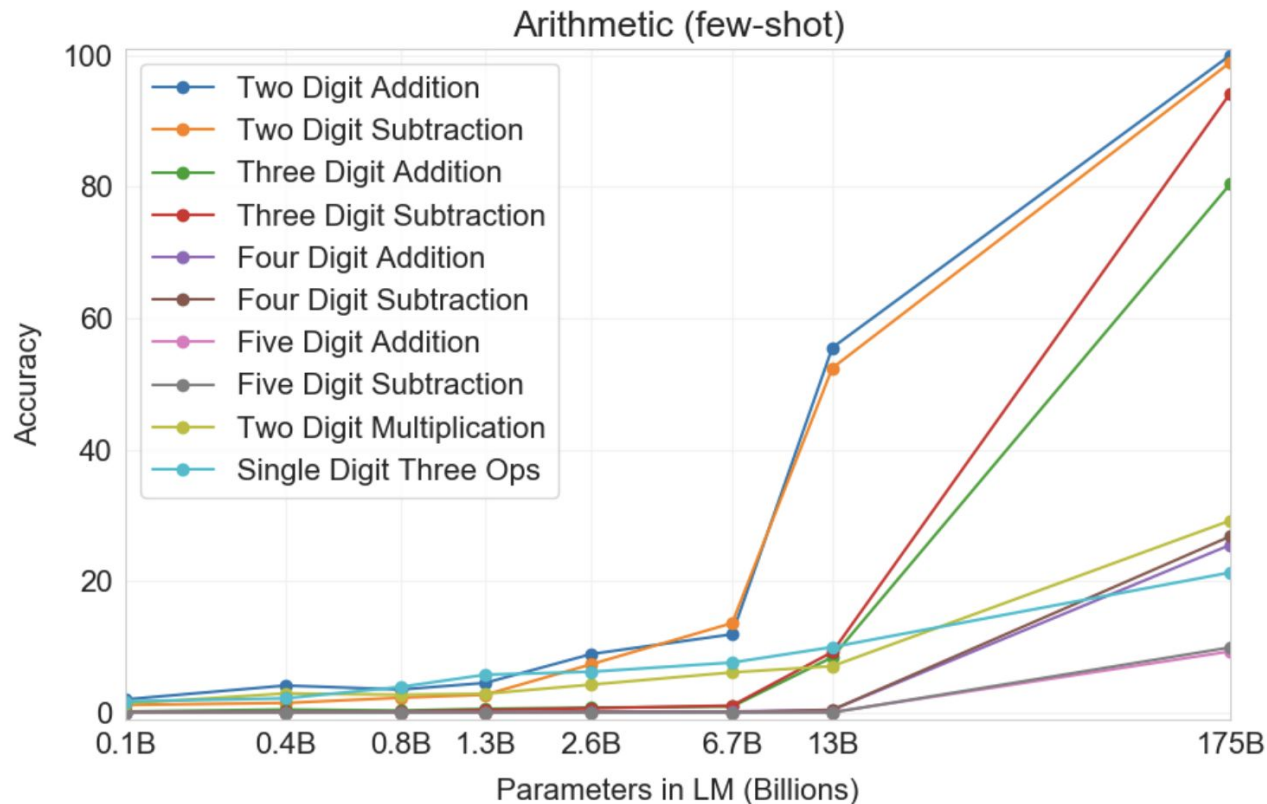
# GPT-3 News Articles

- Humans struggle to identify GPT-3 generated news articles

|                                  | Mean accuracy | 95% Confidence Interval (low, hi) | $t$ compared to control ( $p$ -value) | “I don’t know” assignments |
|----------------------------------|---------------|-----------------------------------|---------------------------------------|----------------------------|
| Control (deliberately bad model) | 86%           | 83%–90%                           | -                                     | 3.6 %                      |
| GPT-3 Small                      | 76%           | 72%–80%                           | 3.9 ( $2e-4$ )                        | 4.9%                       |
| GPT-3 Medium                     | 61%           | 58%–65%                           | 10.3 ( $7e-21$ )                      | 6.0%                       |
| GPT-3 Large                      | 68%           | 64%–72%                           | 7.3 ( $3e-11$ )                       | 8.7%                       |
| GPT-3 XL                         | 62%           | 59%–65%                           | 10.7 ( $1e-19$ )                      | 7.5%                       |
| GPT-3 2.7B                       | 62%           | 58%–65%                           | 10.4 ( $5e-19$ )                      | 7.1%                       |
| GPT-3 6.7B                       | 60%           | 56%–63%                           | 11.2 ( $3e-21$ )                      | 6.2%                       |
| GPT-3 13B                        | 55%           | 52%–58%                           | 15.3 ( $1e-32$ )                      | 7.1%                       |
| GPT-3 175B                       | 52%           | 49%–54%                           | 16.9 ( $1e-34$ )                      | 7.8%                       |

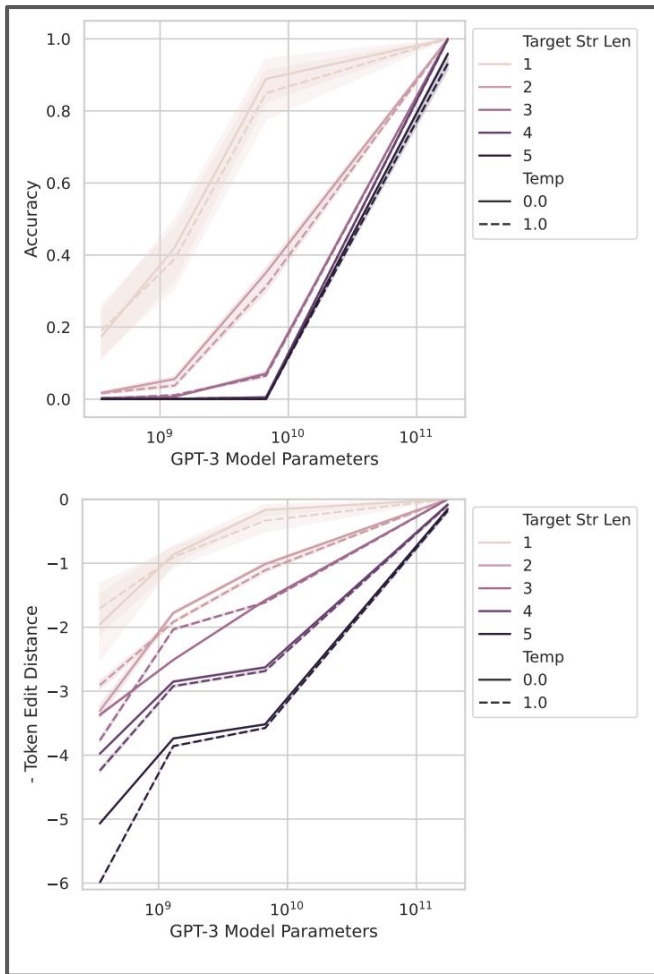
**Table 3.11: Human accuracy in identifying whether short (~200 word) news articles are model generated.**

# Emergent Capabilities



# Is Emergence a Mirage?

- Look at four digit addition under different metrics
  - Emergence can be an artefact of the evaluation metric
- Addition capabilities improve smoothly when using a more granular metric
  - Exact match accuracy -> Token edit distance



# Very Versatile

## Explaining a Joke

I will explain these jokes:

(1) The problem with kleptomaniacs is that they always take things literally.

Explanation: This joke is wordplay. Someone who "takes things literally" is someone who doesn't fully understand social cues and context, which is a negative trait. But the definition of kleptomania is someone who literally takes things.

(2) Always borrow money from a pessimist. They'll never expect it back.

Explanation: Most people expect you to pay them back when you borrow money, however a pessimist is someone who always assumes the worst, so if you borrow money from them, they will expect that you won't pay them back anyways.

**Input:** I was supposed to start writing the paper at 5:00 PM. But then I started playing with this cool new language model for 10 minutes. 10 minutes later, it's suddenly 9:30 PM!

**Model Output:** This joke is about how time flies when you're having fun. The person was supposed to start writing a paper at 5:00 PM, but they started playing with a language model instead. They thought they were only playing with it for 10 minutes, but it was actually 4.5 hours.

**Input:** I think I understand why me and my partner are such a good match. I get excited when I smell food and become super alert when the doorbell rings. My girlfriend Shelley becomes ecstatic when she sees a person who she just saw 5 hours ago.

**Model Output:** This joke is about dogs. Dogs get excited when they smell food and when the doorbell rings. The girlfriend is excited to see someone she just saw 5 hours ago, which is a trait of dogs.

# Any problems with auto-regressive LMs?

---

**Prompt:**

What is the purpose of the list C in the code below?

```
def binomial_coefficient(n, r):  
    C = [0 for i in range(r + 1)];  
    C[0] = 1;  
    for i in range(1, n + 1):  
        j = min(i, r);  
        while j > 0:  
            C[j] += C[j - 1];  
            j -= 1;  
    return C[r]
```

---

**GPT-3 175B completion:**

- A. to store the value of C[0]
  - B. to store the value of C[1]
  - C. to store the value of C[i]
  - D. to store the value of C[i - 1]
-

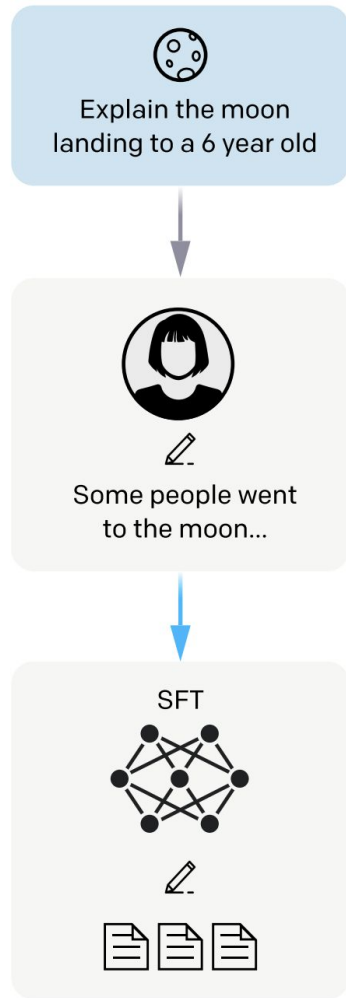
# Instruction Tuning

- Curate a small dataset of instruction following demonstrations
  - Fine-tune the pre-trained LM

A prompt is sampled from our prompt dataset.

A labeler demonstrates the desired output behavior.

This data is used to fine-tune GPT-3 with supervised learning.



# Instruction Tuning

- Aligns LM with user intent!

---

**Prompt:**

What is the purpose of the list C in the code below?

```
def binomial_coefficient(n, r):  
    C = [0 for i in range(r + 1)];  
    C[0] = 1;  
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        j = min(i, r);  
        while j > 0:  
            C[j] += C[j - 1];  
            j -= 1;  
    return C[r]
```

---

**GPT-3 175B completion:**

- A. to store the value of C[0]
- B. to store the value of C[1]
- C. to store the value of C[i]
- D. to store the value of C[i - 1]

---

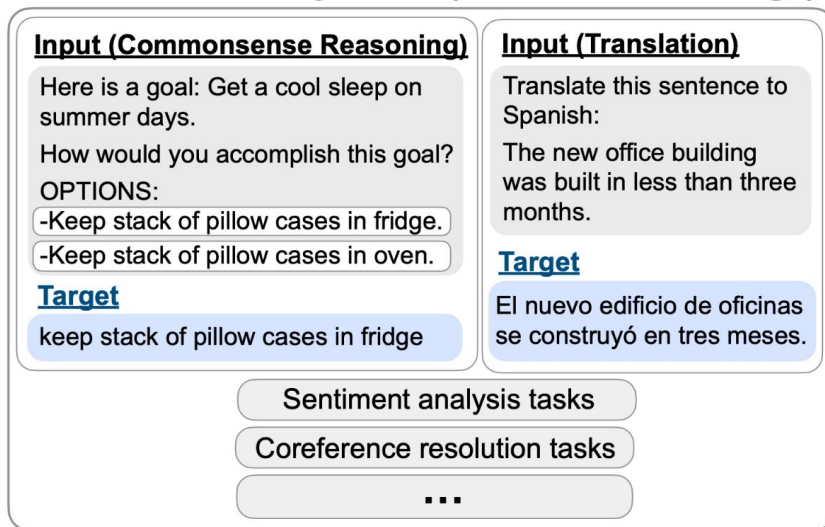
**InstructGPT 175B completion:**

The list C in this code is used to store the values of the binomial coefficient as the function iterates through the values of n and r. It is used to calculate the value of the binomial coefficient for a given value of n and r, and stores the result in the final return value of the function.

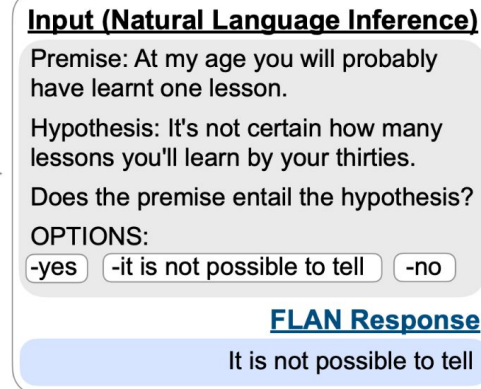
# Instruction Tuning

- Model learns to generalize to unseen instructions at inference-time

## Finetune on many tasks (“instruction-tuning”)



## Inference on unseen task type



# Prompt Engineering

## Standard Prompting

### Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

### Model Output

A: The answer is 27. ❌

## Chain-of-Thought Prompting

### Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls.  $5 + 6 = 11$ . The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

### Model Output

A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had  $23 - 20 = 3$ . They bought 6 more apples, so they have  $3 + 6 = 9$ . The answer is 9. ✅

# Chain of Thought Prompting

- Break down problem into subparts
  - Learns to to it with few-shot in-context learning

## Math Word Problems (free response)

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls.  $5 + 6 = 11$ . The answer is 11.

## Math Word Problems (multiple choice)

Q: How many keystrokes are needed to type the numbers from 1 to 500?  
Answer Choices: (a) 1156 (b) 1392 (c) 1480 (d) 1562 (e) 1788

A: There are 9 one-digit numbers from 1 to 9. There are 90 two-digit numbers from 10 to 99. There are 401 three-digit numbers from 100 to 500.  $9 + 90(2) + 401(3) = 1392$ . The answer is (b).

## CSQA (commonsense)

Q: Sammy wanted to go to where the people were. Where might he go?  
Options: (a) race track (b) populated areas (c) desert (d) apartment (e) roadblock

A: The answer must be a place with a lot of people. Race tracks, desert, apartments, and roadblocks don't have a lot of people, but populated areas do. So the answer is (b).

## StrategyQA

Q: Yes or no: Would a pear sink in water?

A: The density of a pear is about  $0.6 \text{ g/cm}^3$ , which is less than water. Thus, a pear would float. So the answer is no.

## Date Understanding

Q: The concert was scheduled to be on 06/01/1943, but was delayed by one day to today. What is the date 10 days ago in MM/DD/YYYY?

A: One day after 06/01/1943 is 06/02/1943, so today is 06/02/1943. 10 days before today is 05/23/1943. So the answer is 05/23/1943.

## Sports Understanding

Q: Is the following sentence plausible? "Joao Moutinho caught the screen pass in the NFC championship."

A: Joao Moutinho is a soccer player. The NFC championship is part of American football, not soccer. So the answer is no.

## SayCan (Instructing a robot)

Human: How would you bring me something that isn't a fruit?

Explanation: the user wants something to eat that isn't a fruit. An energy bar is not a fruit, so I will bring the user an energy bar.

Plan: 1. find(energy bar) 2. pick(energy bar) 3. find(user) 4. put(energy bar) 5. done().

## Last Letter Concatenation

Q: Take the last letters of the words in "Lady Gaga" and concatenate them.

A: The last letter of "Lady" is "y". The last letter of "Gaga" is "a". Concatenating them is "ya". So the answer is ya.

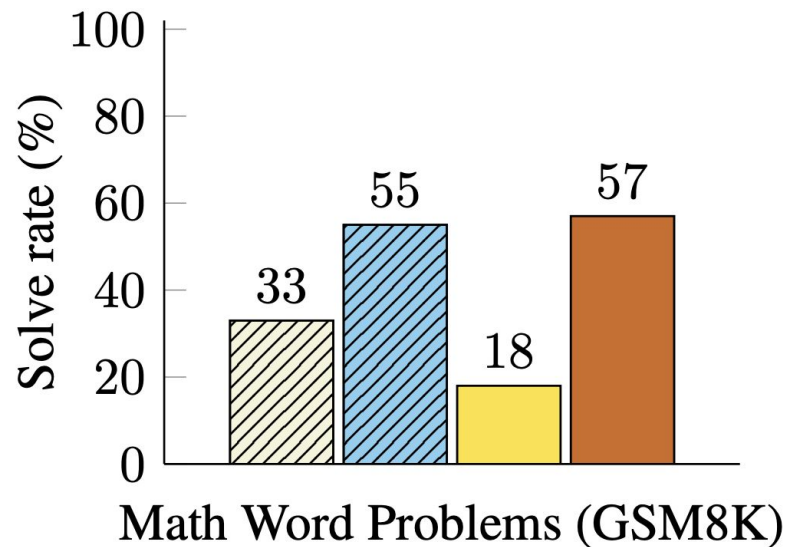
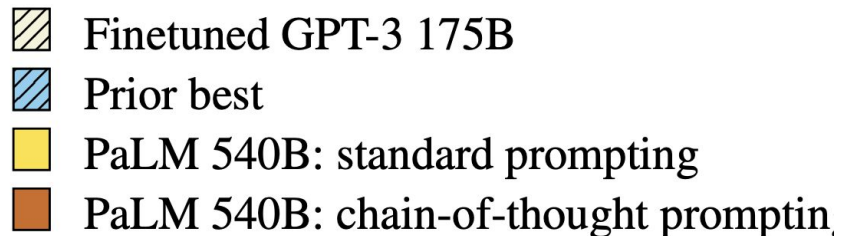
## Coin Flip (state tracking)

Q: A coin is heads up. Maybelle flips the coin. Shalonda does not flip the coin. Is the coin still heads up?

A: The coin was flipped by Maybelle. So the coin was flipped 1 time, which is an odd number. The coin started heads up, so after an odd number of flips, it will be tails up. So the answer is no.

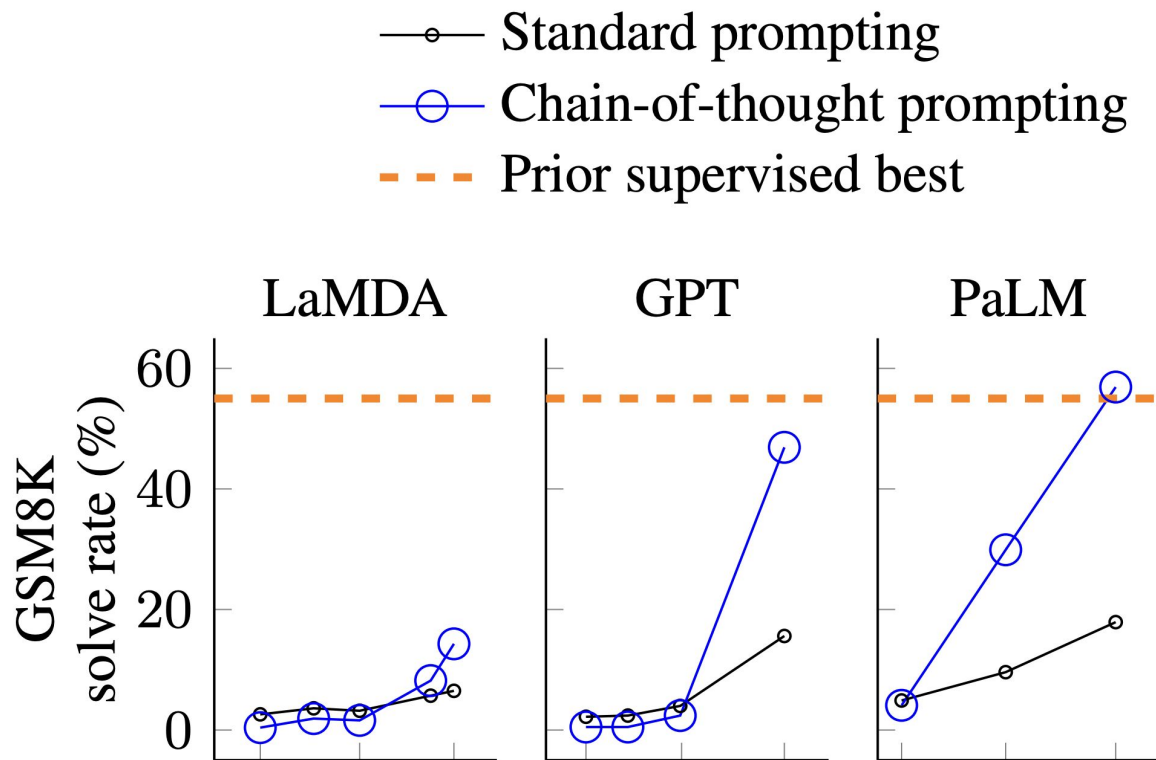
# Chain of Thought Prompting

- Can significantly improve performance



# Chain of Thought Prompting

- No benefit at small scales



# Sensitivity to the Prompt

- Performance depends a lot on the prompt!

Table 4: Robustness study against template measured on the MultiArith dataset with text-davinci-002. (\*1) This template is used in Ahn et al. [2022] where a language model is prompted to generate step-by-step actions given a high-level instruction for controlling robotic actions. (\*2) This template is used in Reynolds and McDonell [2021] but is not quantitatively evaluated.

| No. | Category    | Template                                                  | Accuracy    |
|-----|-------------|-----------------------------------------------------------|-------------|
| 1   | instructive | Let's think step by step.                                 | <b>78.7</b> |
| 2   |             | First, (*1)                                               | 77.3        |
| 3   |             | Let's think about this logically.                         | 74.5        |
| 4   |             | Let's solve this problem by splitting it into steps. (*2) | 72.2        |
| 5   |             | Let's be realistic and think step by step.                | 70.8        |
| 6   |             | Let's think like a detective step by step.                | 70.3        |
| 7   |             | Let's think                                               | 57.5        |
| 8   |             | Before we dive into the answer,                           | 55.7        |
| 9   |             | The answer is after the proof.                            | 45.7        |
| 10  | misleading  | Don't think. Just feel.                                   | 18.8        |
| 11  |             | Let's think step by step but reach an incorrect answer.   | 18.7        |
| 12  |             | Let's count the number of "a" in the question.            | 16.7        |
| 13  |             | By using the fact that the earth is round,                | 9.3         |
| 14  | irrelevant  | By the way, I found a good restaurant nearby.             | 17.5        |
| 15  |             | AbraKadabra!                                              | 15.5        |
| 16  |             | It's a beautiful day.                                     | 13.1        |
| -   |             | (Zero-shot)                                               | 17.7        |

# Why does Chain of Thought Prompting work?

- Breaking problems into steps reduces uncertainty
- Lower uncertainty  $\rightarrow$  lower expected loss
- The training objective rewards this structure

# Why does Chain of Thought Prompting work?

- $47 \times 23 = \dots$ 
  - Could start with 1081
  - Could start with 940
  - Could restate
  - Could explain
  - Could hesitate
  
- Tons of possibilities!

# Why does Chain of Thought Prompting work?

- $47 \times 23$   
=  $47 \times (20 + 3)$   
=  $47 \times 20 + \dots$

## Why does Chain of Thought Prompting work?

- $47 \times 23$   
=  $47 \times (20 + 3)$   
=  $47 \times 20 + 47 \times 3$   
=  $940 + 141$   
=  $1081$
- The next token is much more constrained
- The number of valid continuations collapses

# Why does Chain of Thought Prompting work?

- Decomposing problem into intermediate states where the continuation is obvious!

# Toolformer

- LLMs can be designed to use tools like calculators, QA systems and calendars to produce better results
- LLM are trained to generate and issue queries to external tools

The New England Journal of Medicine is a registered trademark of [QA("Who is the publisher of The New England Journal of Medicine?") → Massachusetts Medical Society] the MMS.

Out of 1400 participants, 400 (or [Calculator(400 / 1400) → 0.29] 29%) passed the test.

The name derives from "la tortuga", the Spanish word for [MT("tortuga") → turtle] turtle.

The Brown Act is California's law [WikiSearch("Brown Act") → The Ralph M. Brown Act is an act of the California State Legislature that guarantees the public's right to attend and participate in meetings of local legislative bodies.] that requires legislative bodies, like city councils, to hold their meetings open to the public.

# Limitations: Hallucinations

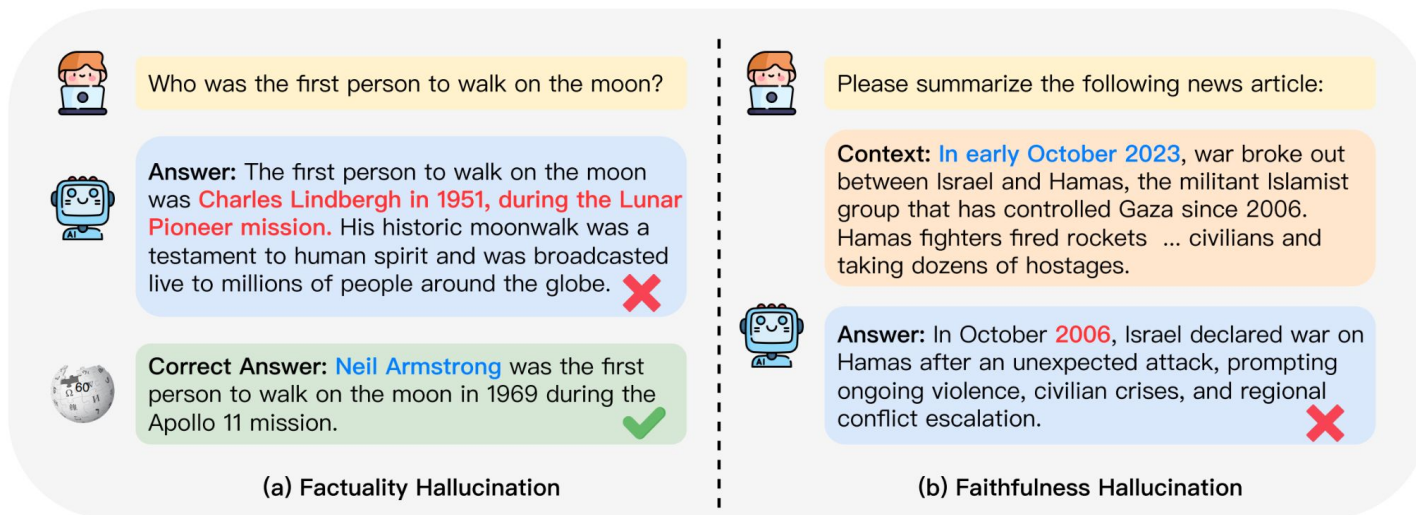


Figure 1: An intuitive example of LLM hallucination.

# Recap

- Different variants of language models
  - Encoder only, encoder-decoder, decoder only
- Performance improves with data and model size
  - Can actually fit power laws and predict performance before training!
- LLMs learn how to do a variety of tasks during pre-training
- A second phase is required to give pre-trained LMs instruction following capabilities
  - Instruction tuning
- Performance can depend on the prompt
- LMs often hallucinate!